BHAGWAN MAHAVIR UNIVERSITY, SURAT BHAGWAN MAHAVIR COLLEGE OF MANAGEMENT MASTER OF COMPUTER APPLICATION (M.C.A.)

A

PROJECT REPORT

ON

“V-CARD”

Submitted in fulfilment towards the award of the degree of

For the Academic Year 2024-2025

Submitted By:

Name & Enrollment Number:

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Internal Guide:

Prof. Khushbhu Patel

Bhagwan Mahavir College of Management

M.C.A.

Surat

Company

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"V-CARD"

was prepared by

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Under my guidance to fulfil the degree of Master of Computer Application (4th semester) of Bhagwan Mahavir University, Surat during the academic year 2024-25.

I certify that this project is their original work and not copied from other sources.

_____
Internal Guide Signature

The Project is forwarded to the University for Further Evaluation.

HOD

The Project is evaluated by:

1. Name & Sign of External Examiner: _____
2. Name & Sign of External Examiner: _____

ACKNOWLEDGEMENT

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Ravva Ajay

M.C.A.

Bhagwan Mahavir College of Management

Surat, 2024-2025

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1 INTRODUCTION

1.1 EXISTING SYSTEM:

- The Current System For Managing Business Contacts Within L&T Involves Manual Data Entry And Storage, Which Is Not Only Inefficient But Also Prone To Errors.
- There Is No Centralized System That Allows For Seamless Digital Storage Of Business Card Details For Employees.
- Time-Consuming: Employees Must Manually Enter Each Contact's Details, Which Can Be Tedious And Repetitive.
- Error-Prone: Manual Data Entry Increases The Risk Of Errors Such As Misspelled Names, Incorrect Phone Numbers, Or Incomplete Data.
- Decentralized Data: There Is No Unified Platform Where All Employees Can Store Or Access Business Contacts. This Results In Fragmented Data Storage Across Devices Or Teams.
- Lack Of Backup And Security: Business Card Information Stored Manually Or In Personal Devices Lacks Secure Backup Options And May Be Lost If Not Properly Maintained.
- Inaccessibility: Information Is Not Always Readily Available Across Platforms Or When Needed, Especially During Remote Work Or While On The Move.

1.2 NEED FOR THE NEW SYSTEM:

- L&T Employees Frequently Interact With Clients, Vendors, And Industry Professionals During Meetings, Conferences, And Networking Events. As A Result, They Accumulate A Large Number Of Business Cards Over Time. However, The Manual Process Of Managing These Cards—Including Storing, Sorting, And Entering Contact Information—Can Be Highly Inefficient And Error-Prone.

KEY DRIVERS FOR THE NEED OF A NEW SYSTEM INCLUDE:

- A Digital Solution To Replace Physical Cards.
- Automated Data Retrieval From The Company Database.
- Ai-Powered Card Scanning For Extracting Details From Physical Cards.

- Integrated Contact Management With Call, Message, And Email Options.
- QR Code & URL Generation For Seamless Sharing.
- LINKEDIN Integration To Enhance Professional Networking

1.3 OBJECTIVE OF THE NEW SYSTEM:

- Develop A Secure, User-Friendly Digital Visiting Card Application For Employees.
- Automate Business Card Data Extraction Using Azure Ai-Powered OCR Scanning.
- Enable Easy Contact Management With Call, Message, And Email Integration.
- Provide Personalized Or Codes And URLs For Seamless Sharing.
- Ensure Data Security And Compliance Through Structured Testing And VAPT Validation.

1.4 PROBLEM DEFINITION:

- Limited Accessibility: Physical Cards Can Be Misplaced Or Lost.
- Manual Data Entry: Contact Details Need To Be Saved Manually.
- Lack Of Integration: No Direct Digital Connectivity To Platforms Like LINKEDIN.
- Environmental Impact: Paper-Based Cards Contribute To Waste.

CORE COMPONENTS/MODULES:

- **Profile Display**
- **Scanner**
- **Login Popup**
- **Explore Menu**
- **QR Code**
- **Admin**
- **Customer**

PROJECT PROFILE:

The V-Card Project Is A Digital Visiting Card Solution Designed To Eliminate The Need For Physical Business Cards. Developed As A Web And Mobile Application, It Leverages .Net, Azure Ai, JAVASCRIPT, HTML, CSS, APIS, And SQL To Create A Seamless Experience For Employees To Manage And Share Their Professional Profiles.

The Project Enables Automatic Data Retrieval From The Company Database, Ai-Powered Business Card Scanning, Contact Management, And Direct Communication Tools (Call, Message, Email Integration). Additionally, It Features Image Cropping, QR Code, And URL Generation For Easy Sharing.

Following The Waterfall Software Development Lifecycle, The Project.

The Structured Approach Ensured A Secure, User-Friendly, And Enterprise-Ready Digital Networking Solution For Employees. T Was Assigned By The It Deputy Gm, Planned By Business Analysts (BA), Developed By The Development Team, Tested By QA, Secured By VAPT, And Deployed By The DEVOPS Team.

1.5 ASSUMPTIONS AND CONSTRAINTS:

❖ ASSUMPTIONS

- User Base: The System Will Be Used Exclusively By L&T Employees For The Purpose Of Scanning And Managing Business Card Data.
- Authentication & Security: The System Will Integrate With L&T's Internal Authentication Mechanisms (E.G., Psno Login And Domain Credentials) To Ensure Secure Access And Data Handling.
- Infrastructure Support: It Is Assumed That L&T's It Infrastructure Will Support The Deployment, Scalability, And Maintenance Of The Application.
- Device Accessibility: Users Will Have Access To Compatible Devices (Desktop/Mobile With Camera Access) Required For Scanning Business Cards.

❖ **Constraints**

- OCR Limitations: The Accuracy Of Optical Character Recognition (OCR) May Vary Based On Factors Such As Card Quality, Font Style, Layout, And Lighting During Scanning.
- Initial Scope: External System Integration (E.G., Third-Party CRMS, Global Directories) Will Not Be Available In The Initial Release And May Be Considered For Future Phases.
- Platform Limitations: The First Release Is Optimized For Web Usage, With Mobile Functionality In Progress Or Limited Depending On Device/Browser Compatibility.

1.6 ADVANTAGES OF THE PROPOSED SYSTEM:

ADVANTAGES:

- Streamlined Contact Management: L&T Employees Can Save Time By Automating The Process Of Storing Business Card Data.
- Secure Access: Contacts Are Stored Securely In The Cloud, Accessible Only By Authenticated L&T Users.
- Efficient Sharing: Users Can Easily Share Contacts Via QR Codes, Urls, Or Within Internal L&T Platforms.
- Organized Database: The System Automatically Categorizes And Stores Contact Details, Making It Easier For Employees To Find And Manage Their Contacts.

1.7 LIMITATIONS OF THE PROPOSED SYSTEM:

LIMITATIONS:

- The System Will Initially Be Restricted To L&T Employees.
- There May Be Occasional Errors In OCR Processing For Poorly Printed Or Handwritten Cards.
- Initial Setup May Require User Training For Optimal Scanning And Usage.

2 REQUIREMENTS GATHERING & TARGETED USERS

2.1 TARGET USERS:

- L&T Employees
- Administrator Of L&T
- Management Of L&T

2.2 REQUIREMENTS:

HARDWARE CONFIGURATION (CLIENT SIDE):

➢ **RAM** : 512 MB

➢ **Hard disk** : 10 GB

➢ **Processor** : 1.0 GHz

HARDWARE CONFIGURATION (SERVER SIDE):

➢ **RAM** : 1 GB

➢ **Hard disk** : 20 GB

➢ **Processor** : 2.0 GHz

SOFTWARE REQUIREMENT (CLIENT SIDE):

➢ **Web Browser** : Chrome, Edge Or Any Compatible Browser

➢ **Operating System** : Windows10 Or Above

SOFTWARE REQUIREMENT (SERVER SIDE):

➢ **Web Server** : Apache

➢ **Server-Side Language** : PHP5.6 Or Above Version

➢ **Database Server** : MYSQL

➢ **Operating System** : Windows 10 Or Above

➢ **Web Browser** : Edge, Brave, CROME, Firefox Or Any Compatible Browser

3 SYSTEM DESIGN

3.1 USE CASE DIAGRAM

ACTORS:

1. **User** – L&T User Can Scan Business Cards
2. **L&T Management (Admin)** – A System Administrator Who Manages The Platform And Users.

USE CASES FOR (FRONT-END INTERACTIONS):

The Use Case Diagram Will Illustrate How L&T Users Interact With The System, Including Tasks Like Scanning Business Cards, Storing Them, And Sharing Them Within The Organization. This Also Includes The Admin Interface For Managing User Access And Permissions.

ACTORS INVOLVED

- **L&T Employee (User):**A Regular Employee Using The V-Card Application For Scanning And Managing Business Cards.
- **Admin:** Responsible For Managing Users, Permissions, Data Integrity, And Overseeing Application Usage.
- **IT Support (Optional):** Handles Technical Issues, Monitors Logs, And Supports System Operations.

PRIMARY USE CASES

Scan Business Card:

- Users Scan Physical Business Cards Using The Mobile/Web App.
- The System Uses AI-Based OCR To Extract Contact Details.

Save Contact Information

- Parsed Data Is Stored Securely In A Personal Or Shared Repository.
- Tags Or Labels Can Be Added For Better Classification.

Edit Extracted Data

- Users Can Review And Edit The AI-Extracted Content For Accuracy Before Saving.

Search Contacts

- Users Can Search Stored Contacts Using Name, Company, Or Tags.

Share Business Card

- Contact Information Can Be Shared Within L&T Using QR Codes, Urls, Or Direct App Sharing.

User Login & Authentication

- Secure Login Via L&T Single Sign-On (SSO) Or Credentials.
- Multi-Factor Authentication Support (If Applicable).

Manage User Roles (Admin)

- Admin Can Assign Roles (User/Admin) And Manage Access Controls.

Monitor Usage (Admin)

- Admin Can View Usage Statistics, Logs, And System Health Reports.

Export Contacts (User/Admin)

- Export Contact Data To CSV, Outlook, Or Other Systems (Based On Permissions).

Delete Contact (User/Admin)

- Users Can Delete Their Saved Contacts; Admins Can Delete Any Data If Required.

Backup & Restore (Admin/IT Support)

- Ensures Contact Data Is Regularly Backed Up And Can Be Restored If Needed.

Report Issue (User)

- Users Can Report Scanning Or App Issues Directly To IT Support.

Manage Tags & Labels

- Allows Categorization Of Contacts Under Departments, Project Names, Etc.

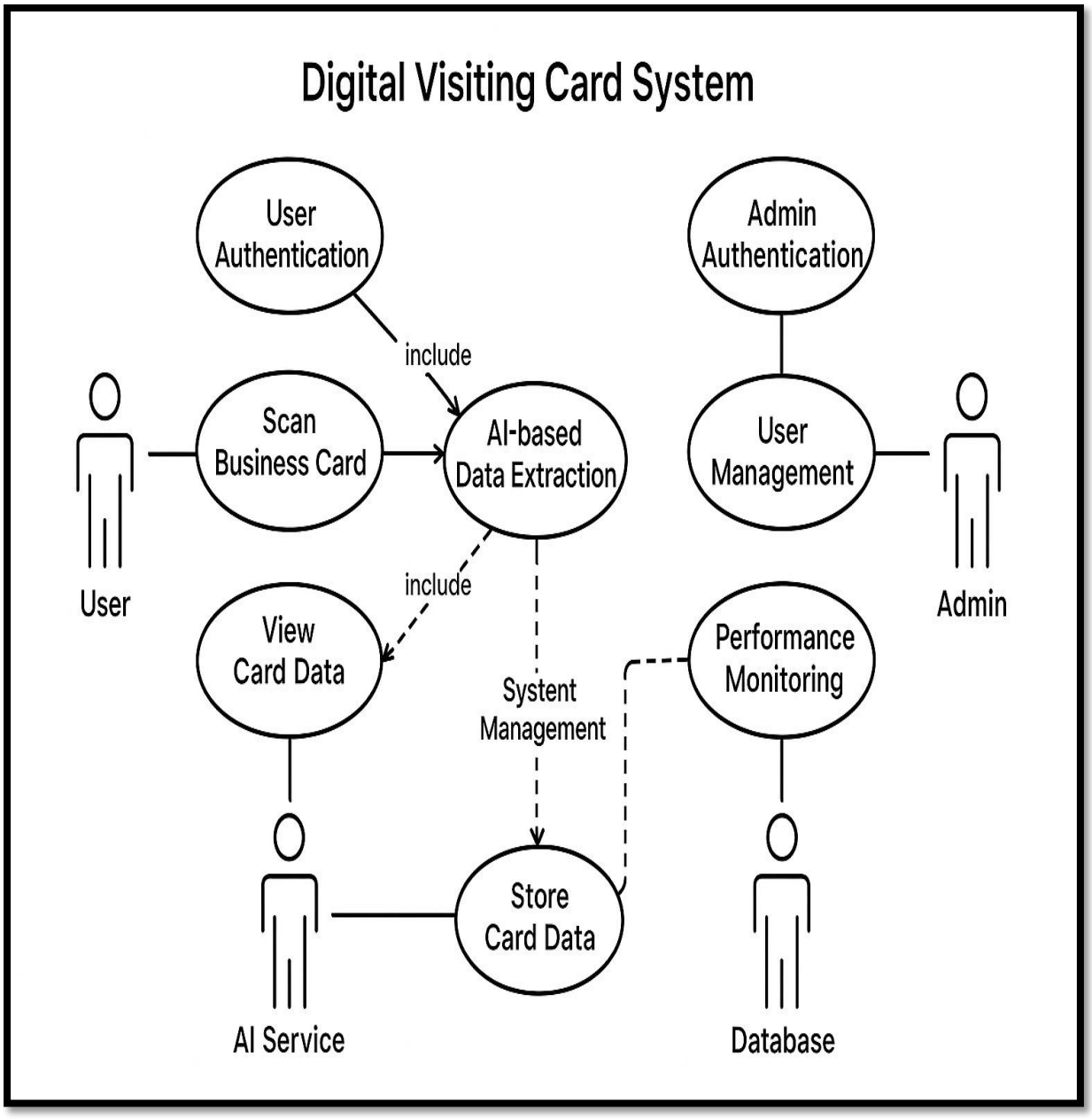
Use Case Diagram Illustration

- (Insert A Use Case Diagram Here Using UML With The Actors And Use Cases Above. You Can Create This Using Tools Like Lucidchart, Draw.Io, Or Visual Paradigm.)

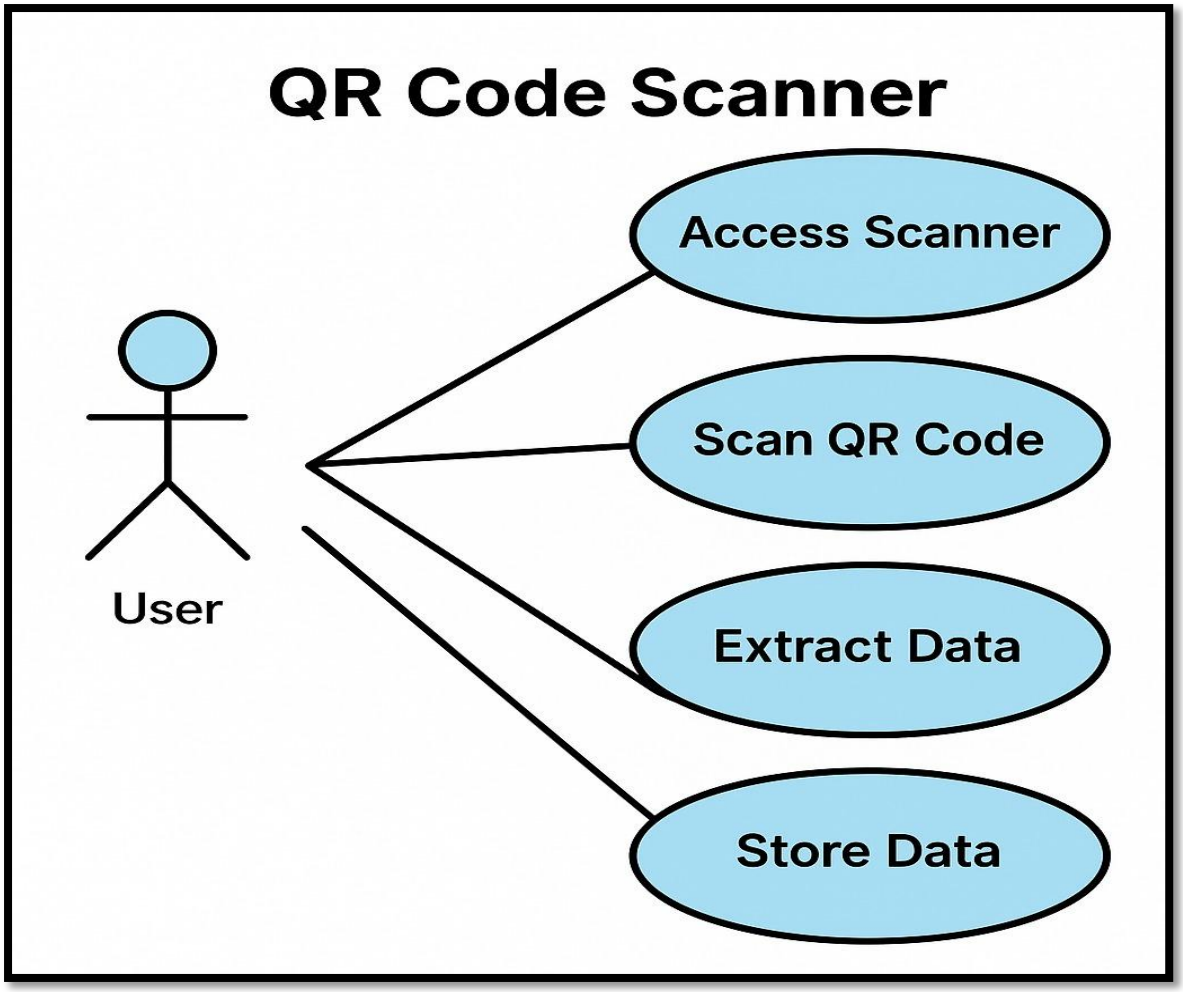
The Diagram Should Show:

- Users Connected To Scan Business Card, Save Contact, Search, Edit, Share, Etc.
- Admin Connected To Manage Roles, Monitor Usage, Backup/Restore, Etc.

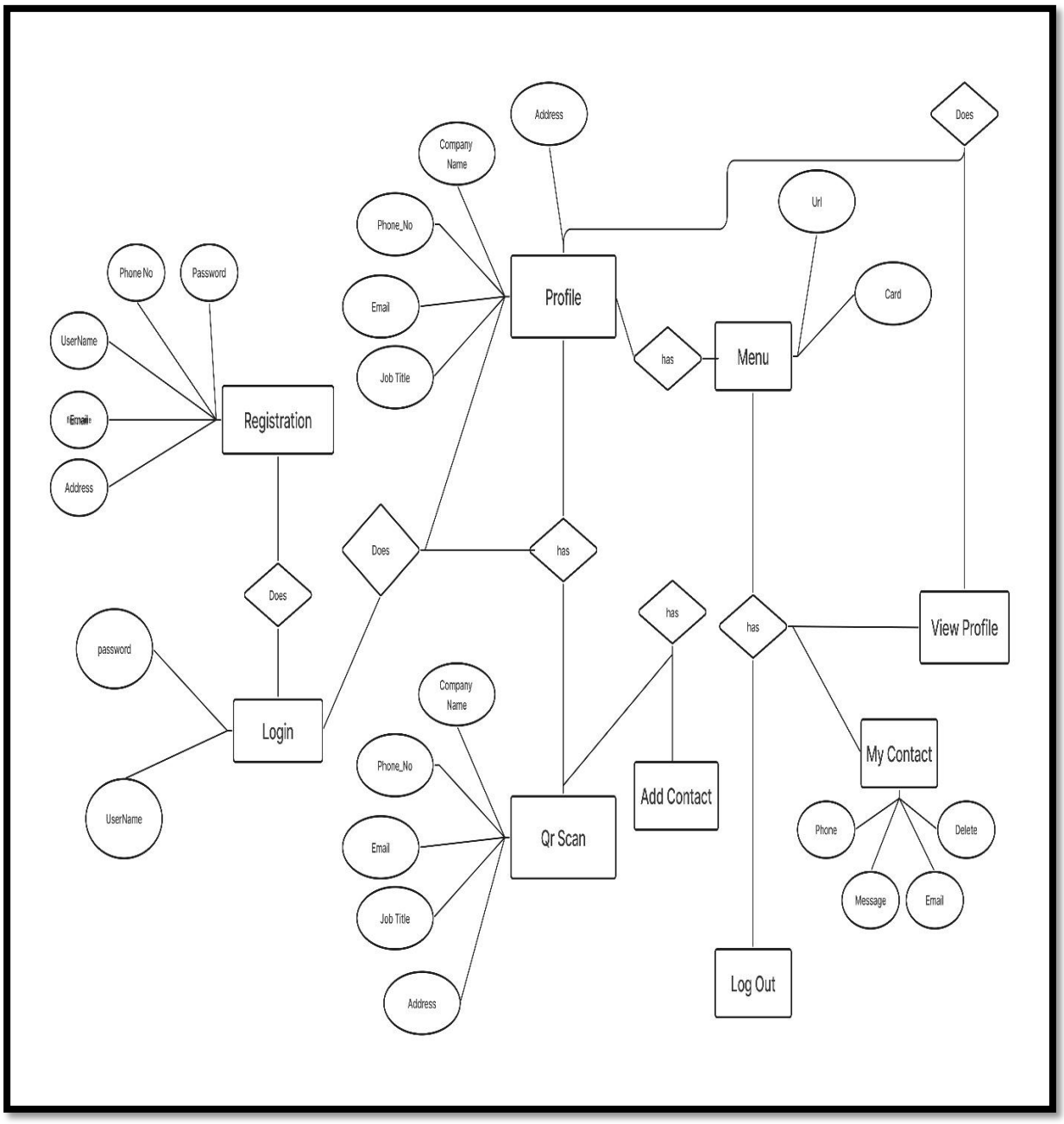
Admin:



Client :

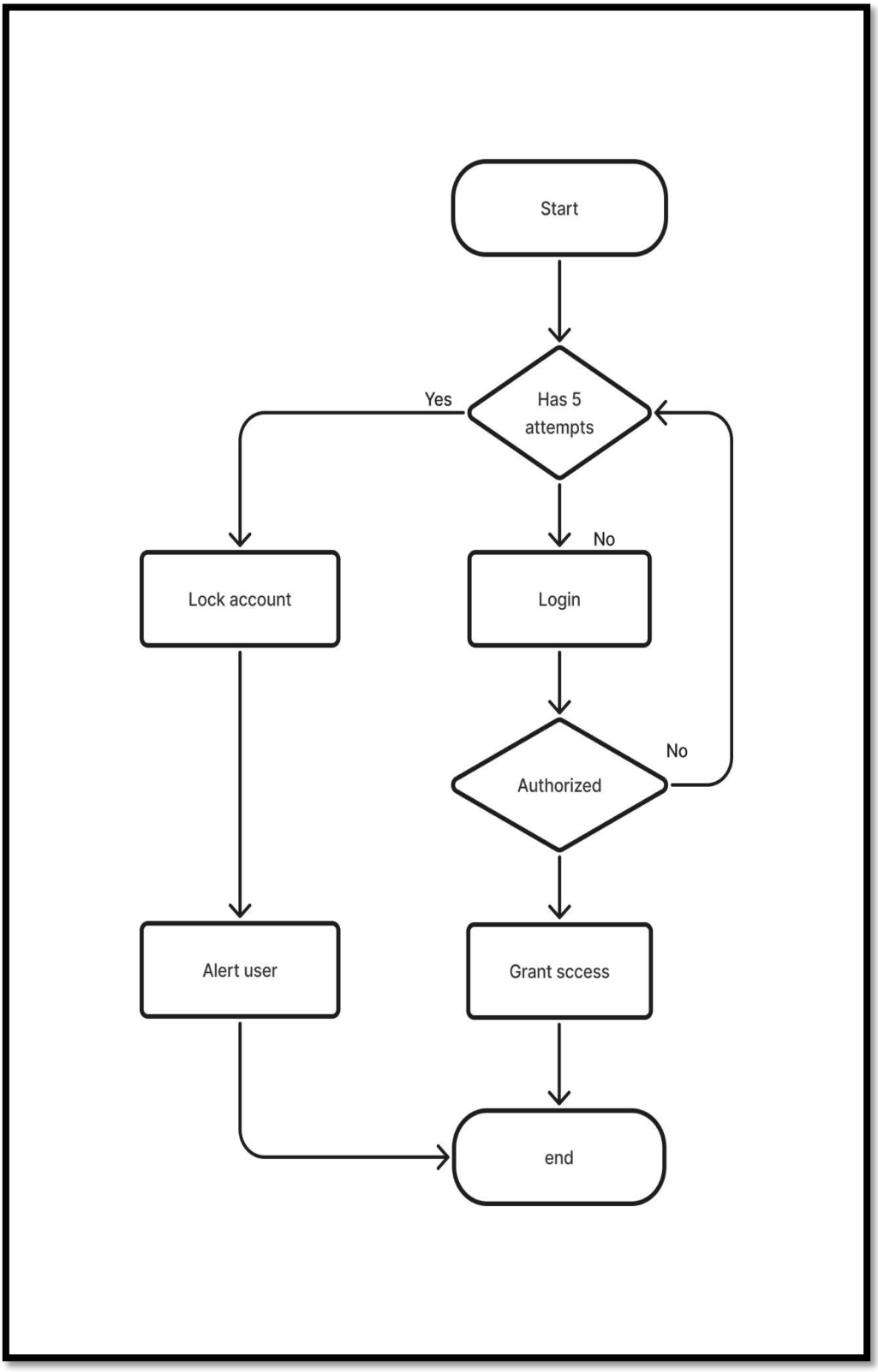

QR Code Scanner
Access Scanner
Scan QR Code
Extract Data
Store Data
User

3.2 ER Diagram

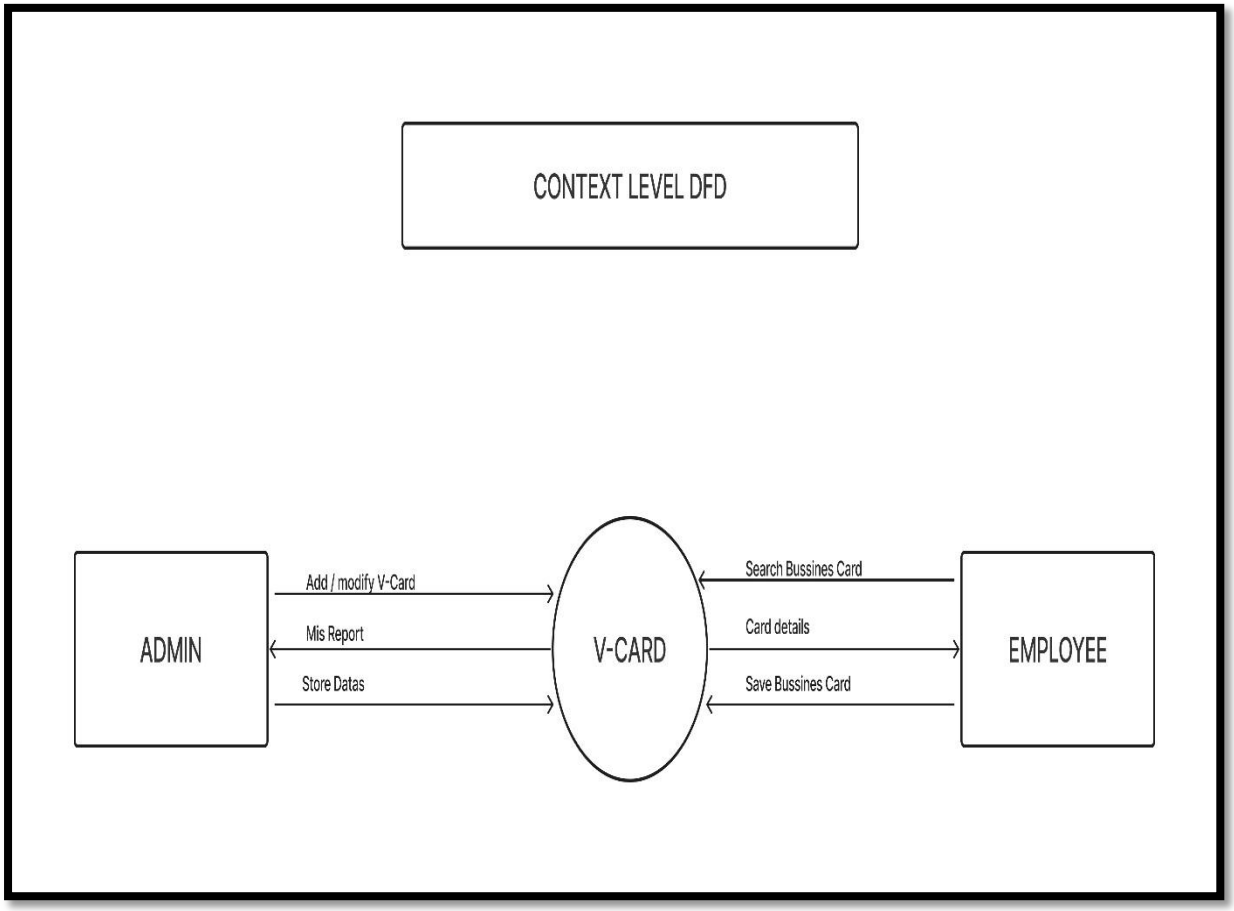


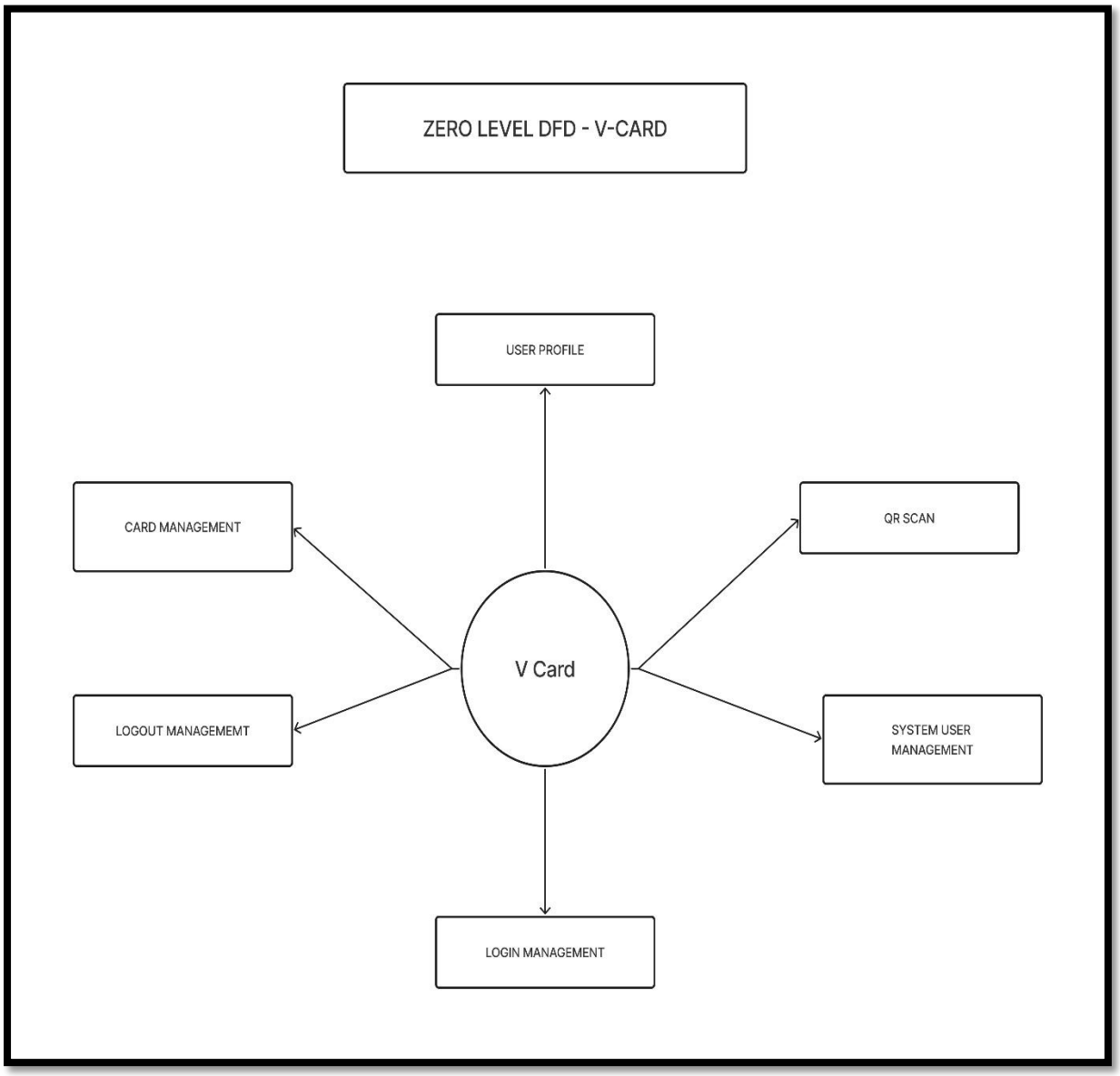
3.3 FLOWCHART(USER LOGIN) & DFD

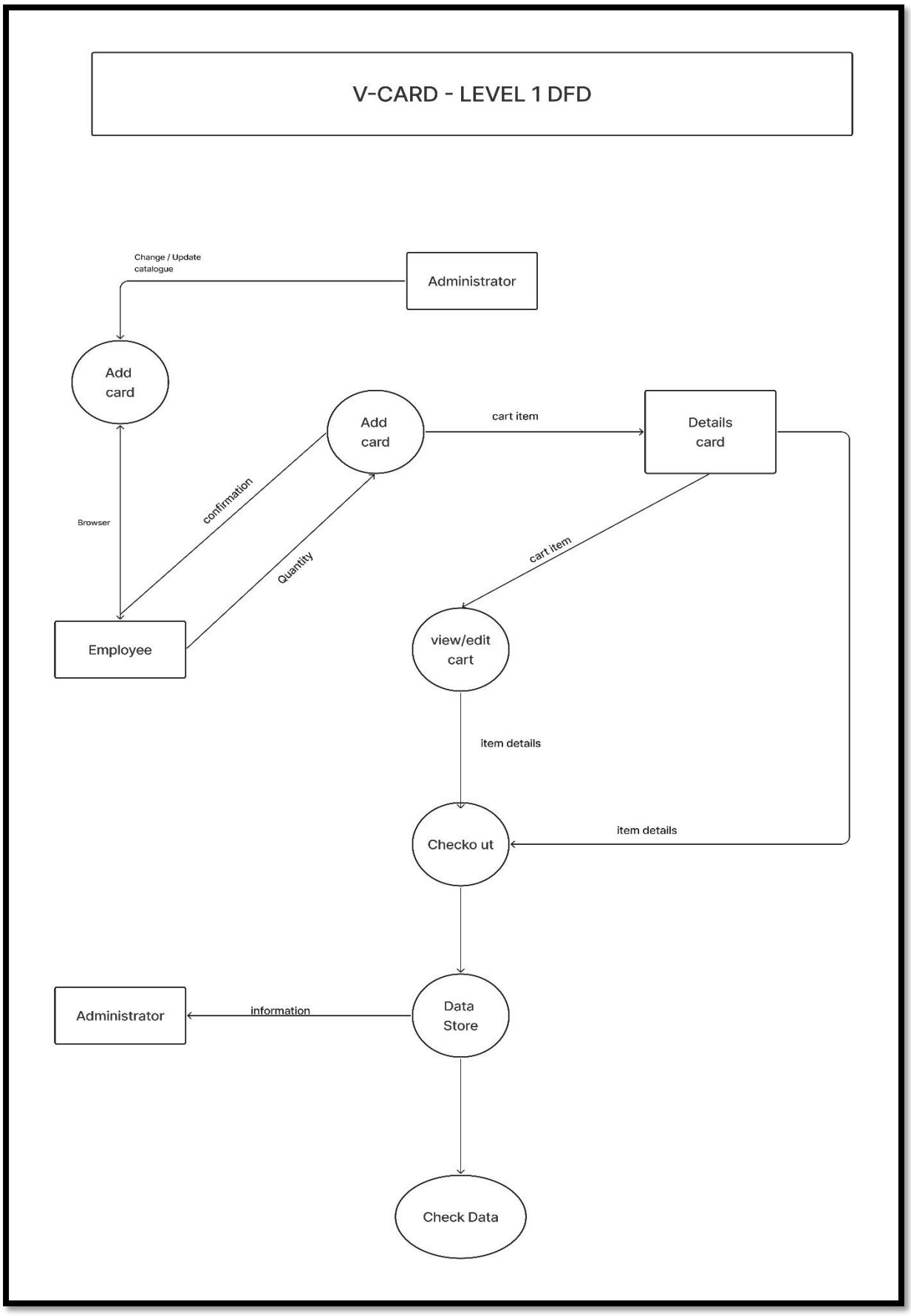
❖ FLOWCHART:

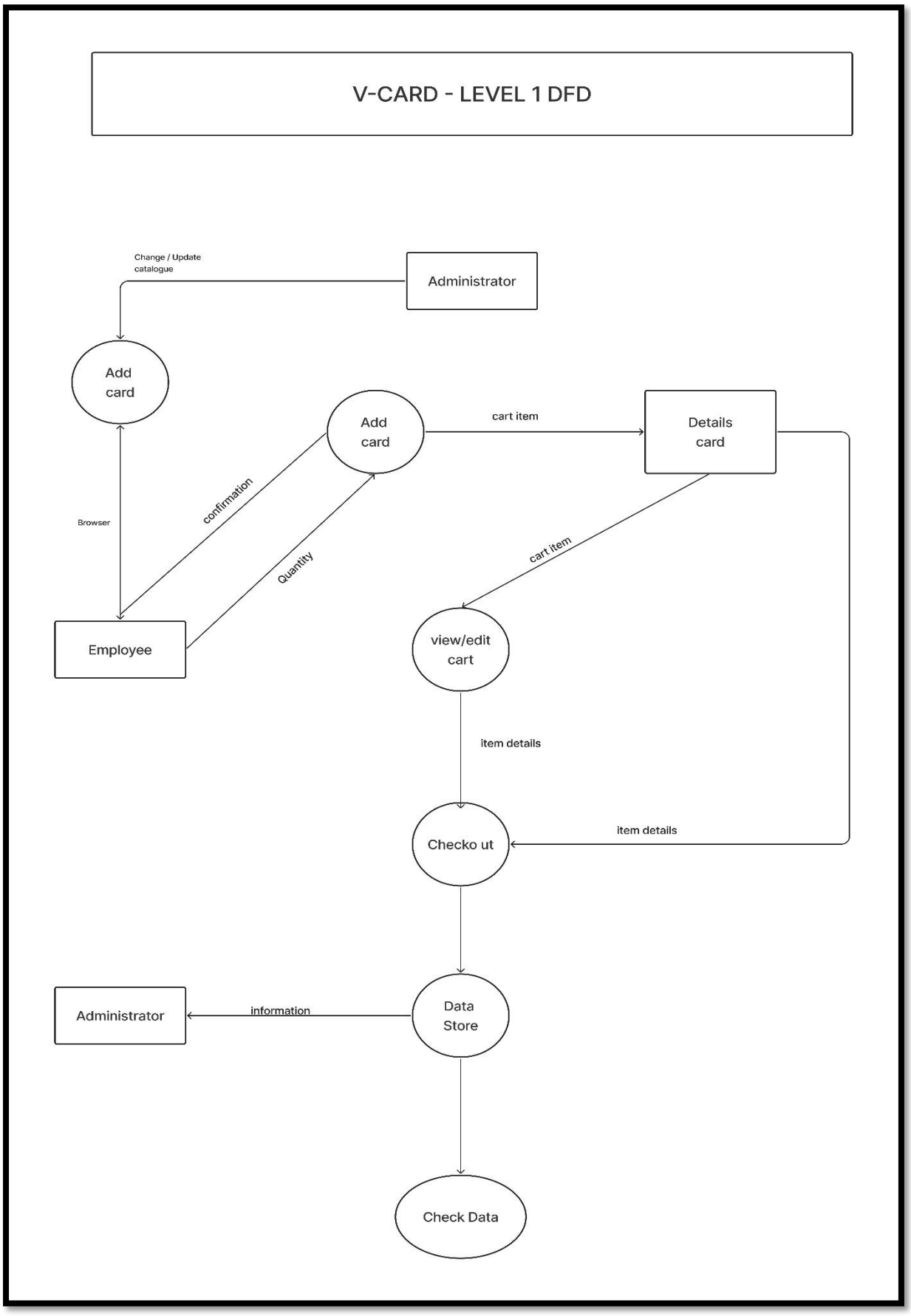


❖ DFD:


CONTEXT LEVEL DFD
ADMIN
Add / modify V-Card
Mis Report
Store Datas
V-CARD
Search Bussines Card
Card details
Save Bussines Card
EMPLOYEE

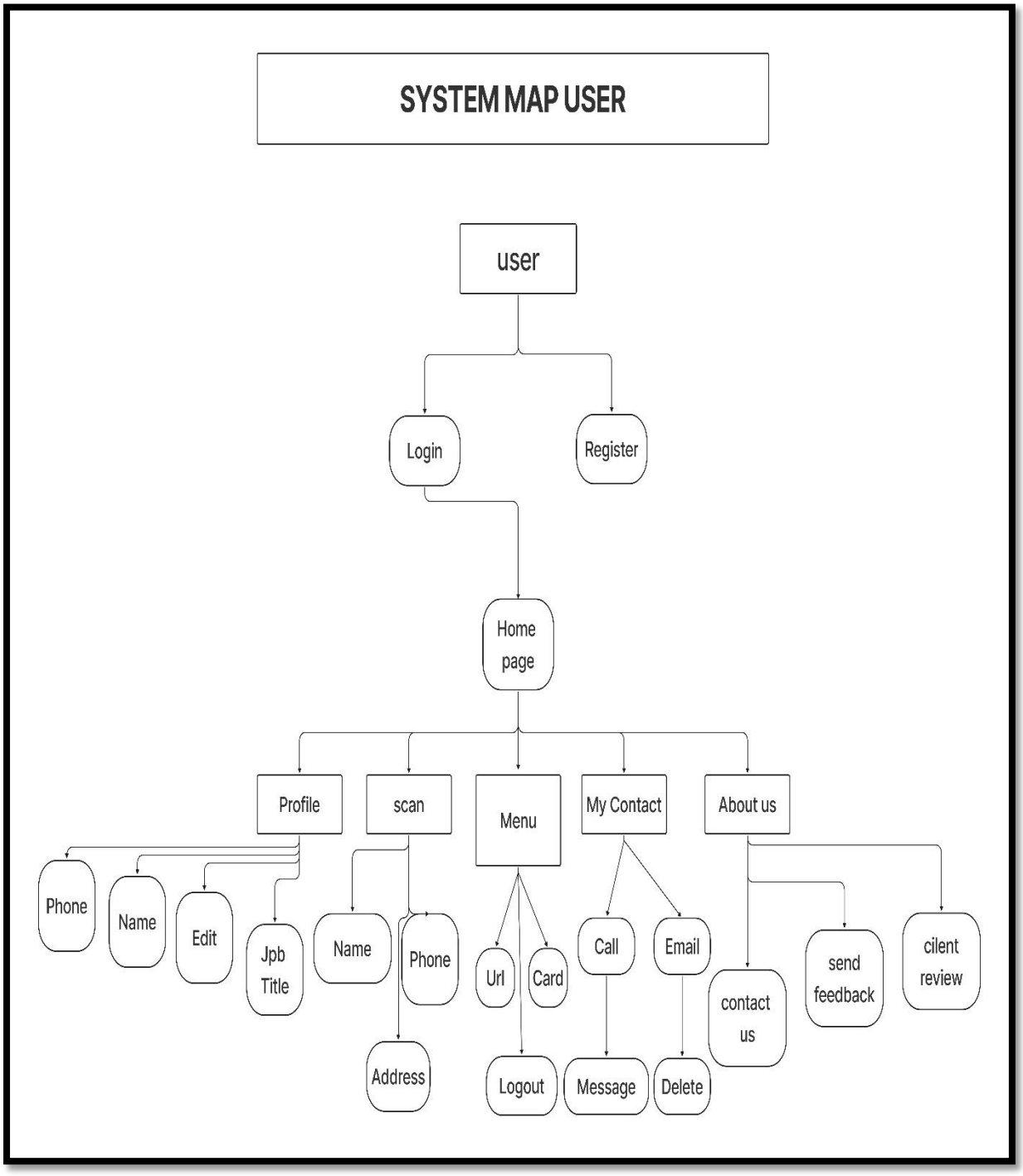

ZERO LEVEL DFD - V-CARD
USER PROFILE
CARD MANAGEMENT
QR SCAN
V Card
LOGOUT MANAGEMEMT
SYSTEM USER MANAGEMENT
LOGIN MANAGEMENT


V-CARD - LEVEL 1 DFD
Change / Update catalogue
Administrator
Add card
Add card
cart item
Details card
Browser
confirmation
Quantity
cart item
Employee
view/edit cart
item details
Checko ut
item details
Administrator
information
Data Store
Check Data

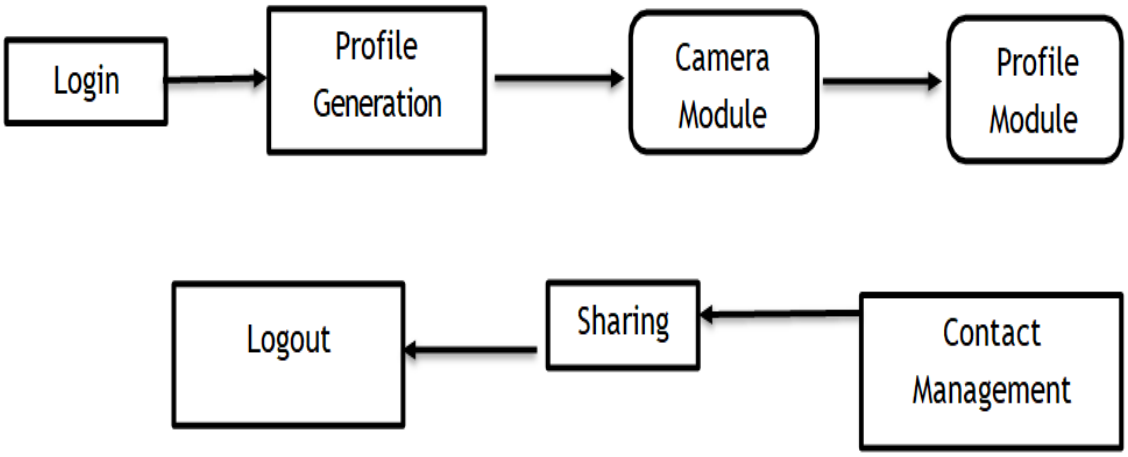

V-CARD - LEVEL 1 DFD
Change / Update catalogue
Administrator
Add card
Add card
cart item
Details card
Browser
confirmation
Quantity
cart item
Employee
view/edit cart
item details
Checko ut
item details
Administrator
information
Data Store
Check Data

3.4 USER DIAGRAM & STATE DIAGRAM

❖ USER DIAGRAM



❖ STATE DIAGRAM



3.5 CLASS DIAGRAM

- The Class Diagram provides a structural view of the V-Card system by illustrating the main classes, their attributes, methods, and relationships. It represents the system architecture in an object-oriented model, giving insights into the static structure of the application.
- This diagram helps in understanding how objects interact within the system, such as scanning a card, processing data via AI, storing it securely, and managing access through L&T's authentication.

RELATIONSHIPS:

User uses → Scanner, OCR Module, Business Card

Business Card is stored in → Contact Repository

User is authenticated by → LT User Authentication

Admin inherits from → User

OCR Module returns data to → Business Card

Notification Service notifies → User

DIAGRAM VISUALIZATION:

(You can represent the above structure in a UML class diagram using tools like Lucid chart, Draw.io, or Star UML)

The diagram should show:

- Inheritance (e.g., Admin ← User)
- Associations (e.g., User ↔ Business Card, User → L&T User Authentication)
- Composition or aggregation if necessary for data management components.

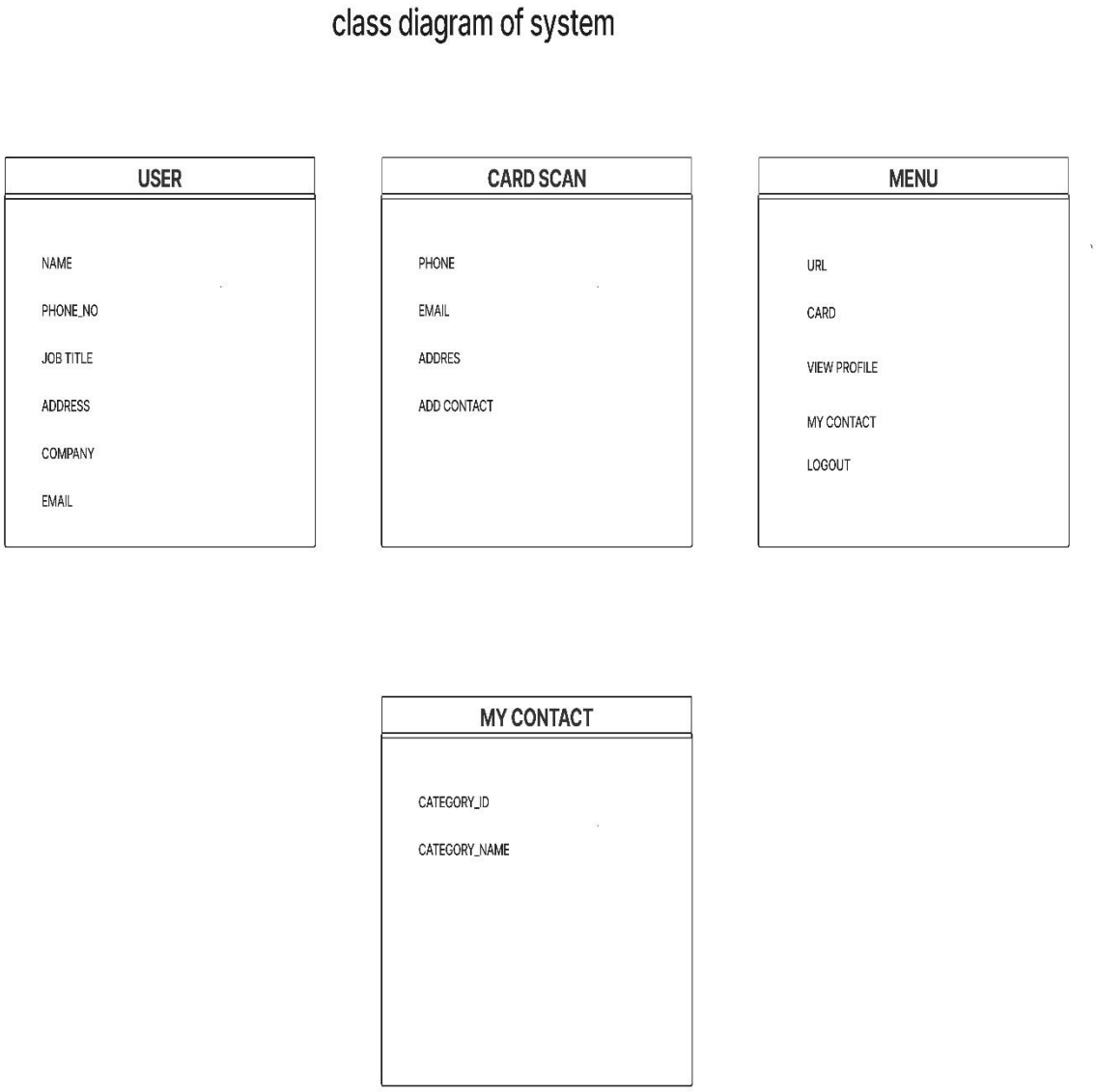
Purpose of the Class Diagram

Visualize object structure and interactions.

Aid in backend/database schema design:

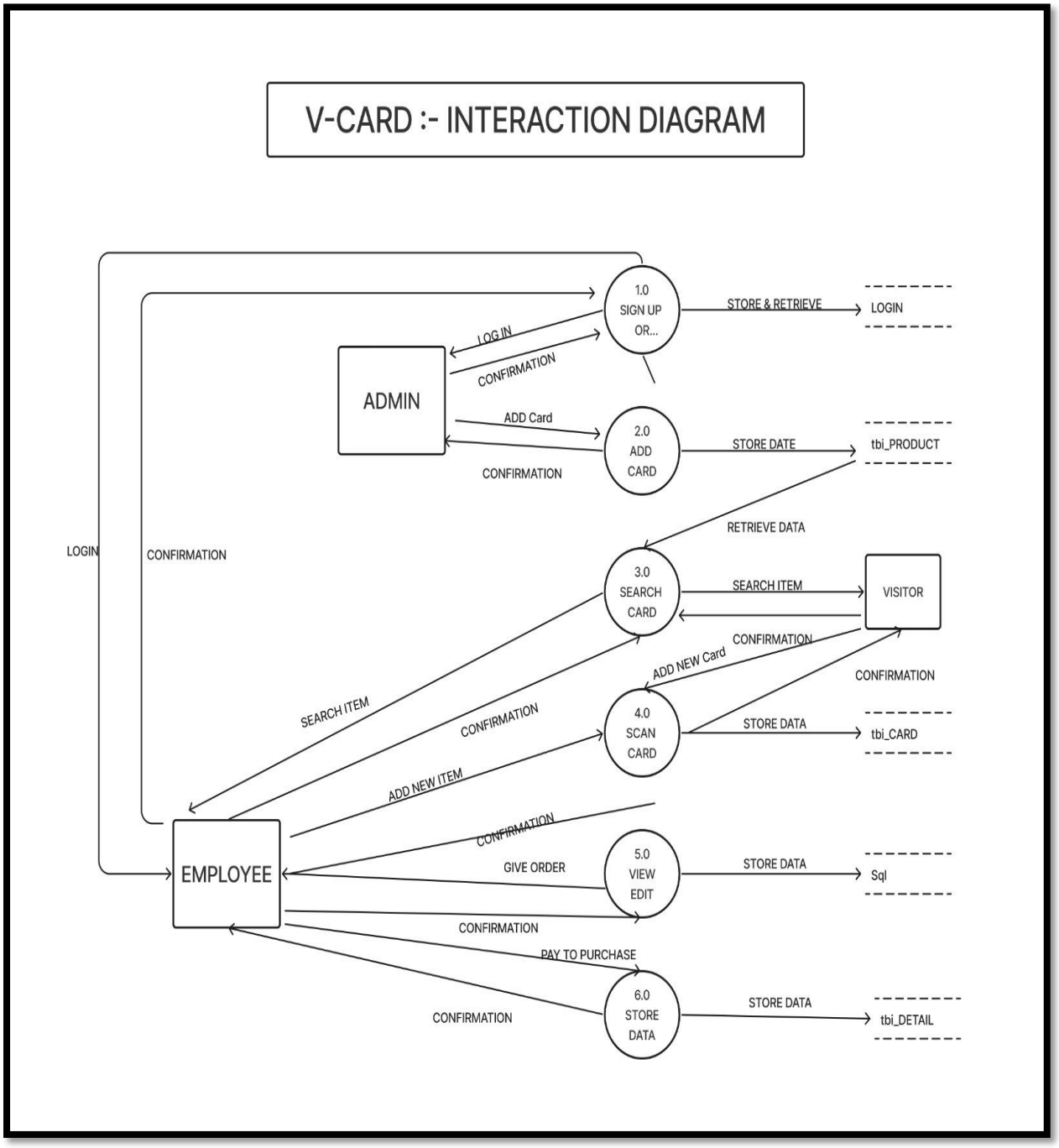
- Improve collaboration among developers and system architects.
- Act as a blueprint for coding the backend and service logic.

3.5 Class Diagram



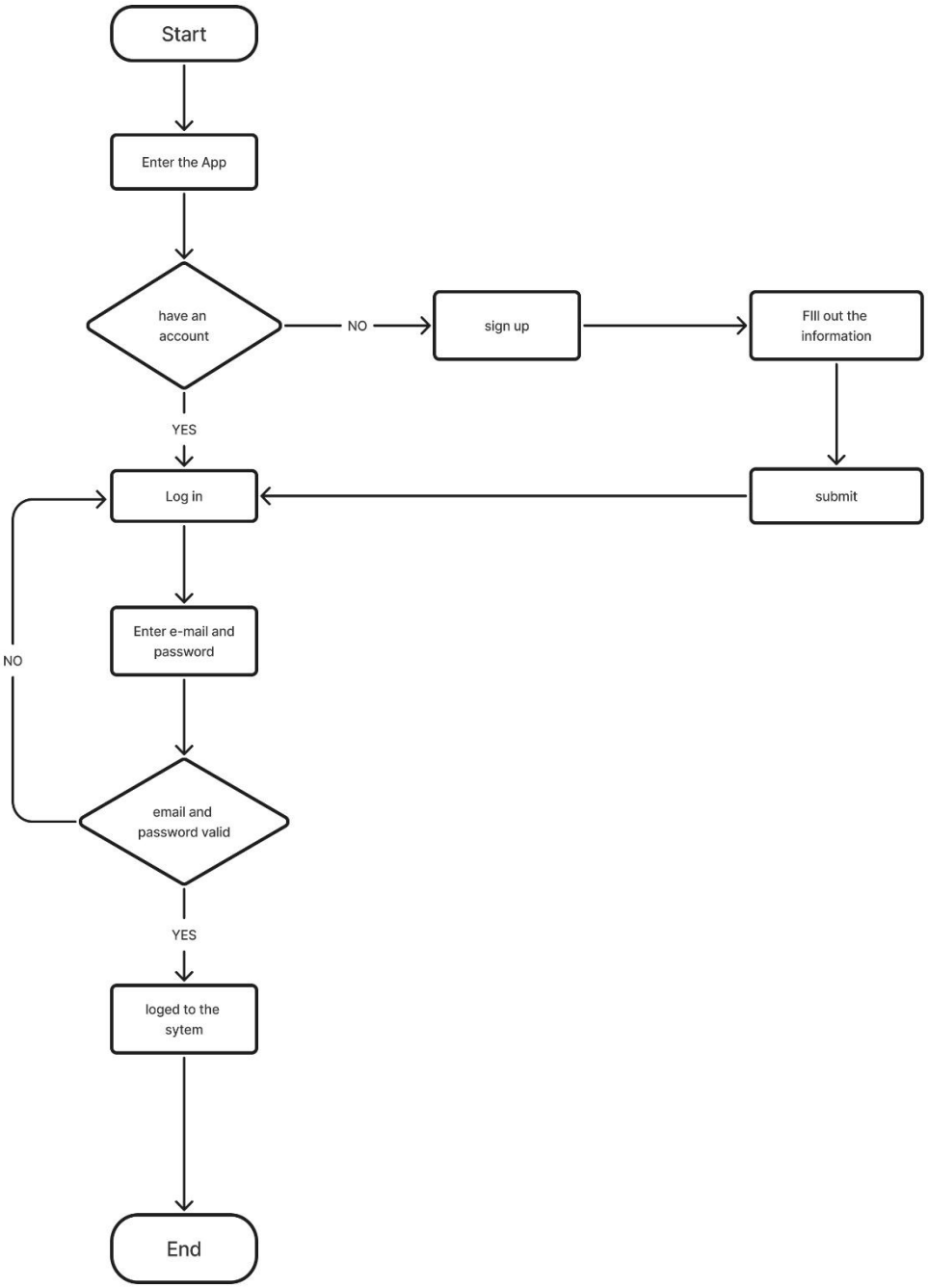
3.6 INTERACTION DIAGRAM

An Interaction Diagram illustrates how objects in your system interact with each other in a specific scenario. It can be represented through sequence diagrams or communication diagram



3.7 ACTIVITY DIAGRAM

➢ An Activity Diagram describes the workflow of a process in the system. For example, the press of make an Appointment.



1) USERS TABLE

Stores customer information (users).

Field	Data Type	Description
User_Id	INT	Unique ID for the user (Primary Key)
First_Name	VARCHAR(100)	User's first name
Last_Name	VARCHAR(100)	User's last name
Email	VARCHAR(255)	User's email address
Password	VARCHAR(255)	Encrypted password
Phone_Number	VARCHAR(15)	User's phone number
Address	TEXT	User's shipping address
Job Title	VARCHAR(255)	Account creation date

1) ADMIN TABLE

Field	Data Type	Description
Admin_id	INT	Unique ID for the user (Primary Key)
Admin_name	VARCHAR(100)	User's first name
email	VARCHAR(255)	User's email address
password	VARCHAR(255)	Encrypted password

4 DEVELOPMENT

4.1 CODING STANDARDS

The V-Card Project Will Follow L&T's Internal Development Guidelines And Industry Best Practices To Ensure Clean, Maintainable, And Secure Code Throughout The Software Development Lifecycle.

Standard Area	Description
Modular Coding	Code Will Be Divided Into Reusable And Maintainable Modules, Following SOLID Principles To Ensure Separation Of Concerns And Scalability.
Naming Conventions	- Variables, Functions: CAMELCASE - Classes, Interfaces: PASCALCASE - Constants: UPPER_SNAKE_CASE
Code Formatting	Indentation, Spacing, And Brace Placement Will Be Consistent Using Automated Linters And Formatters (E.G., EDITORCONFIG, STYLECOP).
Commenting & Documentation	Proper Inline Comments For Complex Logic. XML/Swagger Documentation For Apis. Summary Tags For All Public Classes And Methods.
Security Practices	Input Validation, Output Encoding, And Secure Storage Of Credentials. Use Of HTTPS, Parameterized Queries, And Secure Authentication Flows.
Authentication Integration	Integration With L&T's SSO System And Azure Active Directory (AD) For Secure User Management.
Error Handling	Try-Catch Blocks With Custom Error Messages. Global Error Handler In The API Layer. All Exceptions Logged Securely.

Logging	Logging Using SERILOG/AZURE Monitor. Logs Will Include Timestamps, Severity Levels, And User Context When Necessary (Excluding Sensitive Information).
Version Control	Source Control Via Git Using A Branching Strategy (E.G., GITFLOW). Code Reviews Via Pull Requests Will Be Mandatory.
Testing	Minimum 80% Unit Test Coverage Using XUNIT/NUNIT. Integration Tests And Load Tests For Critical Services.
Code Reviews	All Code Changes Will Undergo Peer Review To Maintain Quality And Consistency.

5 AGILE DOCUMENTATION

5.1 AGILE PROJECT CHARTER

VISION

The V-Card Project Is A Digital Visiting Card Solution Designed To Eliminate The Need For Physical Business Cards. Developed As A Web And Mobile Application, It Leverages .Net, Azure Ai, JavaScript, Html, CSS, APIs, And SQL To Create A Seamless Experience For Employees To Manage And Share Their Professional Profiles.

The Project Enables Automatic Data Retrieval From The Company Database, Ai-Powered Business Card Scanning, Contact Management, And Direct Communication Tools (Call, Message, Email Integration). Additionally, It Features Image Cropping, QR Code, And URL Generation For Easy Sharing.

Following The Waterfall Software Development Lifecycle, The Project.

The Structured Approach Ensured A Secure, User-Friendly, And Enterprise-Ready Digital Networking Solution For Employees. T Was Assigned By THE It Deputy Gm, Planned By Business Analysts (BA), Developed By The Development Team, Tested By QA, Secured By VAPT, And Deployed By The DEVOPS Team.

OBJECTIVES

- Develop A Secure, User-Friendly Digital Visiting Card Application For Employees.
- Automate Business Card Data Extraction Using Azure Ai-Powered OCR Scanning.
- Enable Easy Contact Management With Call, Message, And Email Integration.
- Provide Personalized OR Codes And URLS For Seamless Sharing.
- Ensure Data Security And Compliance Through Structured Testing And VAPT Validation.

SCOPE OF THE PROJECT

● Digital Visiting Card Creation With Auto-Filled Employee Details From The Company Database.

● Ai-Powered Card Scanning For Data Extraction From Physical Business Cards.

● Contact Management With One-Click Call, Message, And Email Functionalities.

● Image Cropping For Adjusting Scanned Cards Before Saving.

● QR Code And URL Generation For Easy Sharing.

PURPOSE & COMPARISION

❖ PURPOSE

The Primary Purpose Of The V-card Project Is To Provide Employees With A Digital, Easily Shareable, And Globally Accessible Business Profile, Eliminating The Need For Carrying Physical Business Cards. It Enhances Networking, Simplifies Contact Management, And Integrates With Modern Ai-Based Data Extraction To Streamline The Process Of Creating And Updating Business Profiles.

❖ COMPARISION

● Traditional Vs. Digital Cards: Eliminates The Need For Paper Cards, Making It Cost-effective And Eco-Friendly.

● Manual Vs. Ai Data Entry: Automated OCR Scanning Significantly Reduces Manual Effort And Errors.

● Physical Exchange Vs. Digital Sharing: QR Code And LINKEDIN Integration Improve Accessibility And Networking.

● Security Improvements: The System Is More Secure With Authentication And Encryption Compared To Traditional Business Card Storage.

PROJECT AND JUSTIFICATION

● The Project Follows The Waterfall Model, Ensuring A Structured And Stepwise Development Approach. This Model Was Chosen Because:

● It Provides A Clear Sequence Of Phases, Reducing Ambiguity.

● Each Module (Authentication, Ai Scanning, Database, Etc.) Undergoes Thorough Validation Before Proceeding.

● It Ensures High-Quality End-Product Delivery, Meeting Enterprise Standards.

5.2 AGILE ROADMAP / SCHEDULE

Phase	Duration	Tasks
1. Planning & Research	Jan 16 – Jan 25	- Requirement gathering - Define core features - Database schema
2. UI/UX Development	Jan 27 - Feb 20	- Develop frontend (HTML ,C and java Script) Components - Home screen, Card search
3. Core Setup	Feb 21 – Mar 1	- Authentication
4. Card System	Mar 2 – Apr 5	- Azure API and Data
5. Backend Development	Apr 6 – Apr 10	- Backend setup - User Authentication
6. Camera Module	Apr 11 – Apr 15	- Capture Physical Cards For Ai-Based Data Extraction
7. Final Release	Apr 16 – Apr 28	- Full release post-feedback implementation

5.3 AGILE PROJECT PLAN

Project Execution Plan (Phase-Based):

Phase	Estimated Effort (Days)	Resources Used
Requirement Analysis	5	BA, Developers
UI/UX Design	8	FIGMA, Developers
Backend & Database Development	16	.Net, SQL Developers
Ai Integration	10	Azure Ai, API Developers
Testing & Debugging	11	QA Team, Selenium
Security Validation	7	VAPT Team, Nessus
Deployment & Publishing	5	DEVOPS Team

5.4 AGILE DEVELOPMENT TOOLS

- **HTML/CSS JAVASCRIPT** - Frontend Development
- **ASP.NET CORE** - Backend development
- **Operating System** - Windows 10 or above
- **Code Management** - Git Hub
- **DATABASE SERVER** - MYSQL

5.5 RISK MANAGEMENT & CONTINGENCY PLAN

This section outlines the potential risks identified in the V-Card system development for L&T users, along with the corresponding mitigation strategies to ensure business continuity and system reliability.

Potential Risk	Mitigation Strategy
Server downtime or Azure service interruption	Use Azure's high-availability features and geo-redundant backup services
OCR/AI extraction errors	Implement manual correction interface and AI model training with feedback loop
Poor user adoption among L&T staff	Conduct on boarding sessions and provide intuitive UI/UX with tooltips/help guides
Data privacy concerns (scanned business card info)	Encrypt sensitive data at rest and in transit; comply with L&T's data governance policy
Unauthorized access to V-Card data	Integrate with L&T's internal SSO and role-based access control
QR/URL sharing misuse	Implement access logs and sharing expiration policies
Data loss or corruption	Use Azure SQL backups and automated daily snapshots

5.6 AGILE USER STORIES

L&T Employees Frequently Interact With Colleagues, Partners, And Clients, Exchanging Physical Business Cards. A Digital Solution Would Make It Easier To Manage And Access These Contacts On-Demand. The System Will Allow Users To Scan Business Cards, Automatically Extract Relevant Contact Details, Store Them Securely, And Provide Functionalities To Share And Edit Those Details Easily Within The Organization.

User Stories For Customers:

Scanning Feature:

The System Must Support The Scanning Of Business Cards Using Both Mobile And Web Interfaces. The User Should Be Able To Capture High-Quality Images For Accurate Data Extraction.

Data Extraction:

The System Should Utilize Optical Character Recognition (OCR) Or AI-Powered Recognition To Automatically Extract Contact Information From Business Cards (E.G., Name, Phone Number, Email, Company, Job Title).

Data Storage:

The Extracted Information Should Be Stored In A Secure Database. Only Authorized Users Within The Organization Should Have Access To The Stored Data.

Contact Management:

Users Should Be Able To View, Edit, Or Delete The Stored Contacts As Needed.

The System Should Allow The Addition Of Custom Tags For Better Organization (E.G., Client, Partner, Colleague).

Search Functionality Should Be Included To Quickly Find Specific Contacts Based On Any Stored Information (E.G., Name, Email, Company).

Sharing Features:

Users Should Be Able To Share Contact Details Easily Through QR Codes Or Urls, Which Can Be Accessed By Other Users In The Organization.

The System Should Allow Users To Generate A Shareable Link For Each Contact Card That Can Be Shared Via Email Or Messaging Platforms.

Security And Privacy:

The System Must Comply With Corporate Data Security Policies, Ensuring All Personal Information Is Encrypted And Access-Controlled.

There Should Be An Audit Log Of Who Accessed, Modified, Or Shared The Contact Details.

User Authentication And Authorization Mechanisms Should Be In Place (E.G., Single Sign-On, Multi-Factor Authentication).

- **User Roles**

Employee (End User):

The primary user who scans and manages business cards.

Can view, edit, and share contact details.

Has access to their own contact information and shared contacts.

Admin:

Manages access permissions and controls user roles.

Oversees security measures and ensures compliance with organizational policies.

Support Team:

Provides assistance to users for any issues related to scanning, data extraction, or contact sharing.

5.7 AGILE SPRINT BACKLOG (PHASE-BASED)

Phase	Tasks (Backlog Items)
Sprint 1	- Define project scope & key features - Design Splash Screen, Login & Signup UI - Design Home Screen, Navigation Bar, Profile Page UI
Sprint 2	- UI Components (Profile List, Scan Card, Checkout Card)
Sprint 3	- Develop frontend (HTML,CSS And JAVASCRIPT)
Sprint 4	- Develop Scan Cart Management System - Implement Scan Card Placement Flow
Sprint 5	Develop Admin Dashboard (Scan, User, Profile, Complaints Management) - Implement Admin Controls (Add/Edit/Delete Profile, Manage Users & Delivery Personnel) - Create Card Status Update Feature

5.8 AGILE TEST PLAN

- **Plan The Testing Follows A Structured Approach Aligned With The Software Development Life Cycle (SDLC) And Focuses On The Following Areas:**

Testing Areas:

- **UI Testing –**

  Verifies Design Consistency And Responsiveness.

- **Backend Testing –**

  Checks Database Operations, API Responses, And System Logic.

- **Validation & Field-Level Checks –**

  Ensures Correct Input Handling.

- **Functional Testing –**

  Confirms That Each Feature Works As Intended.

- **Performance Testing –**

  Assesses System Speed, Load Handling, And Stress Conditions.

- **Security Testing –**

  Identifies Vulnerabilities And Ensures Data Protection.

Testing Types Used:

- **Manual Testing –**

  Performed By Testers For UI, Backend, And Validation Checks.

- **Automated Testing –**

  Conducted Using Tools Like Selenium, Playwright, Postman, Jira, And JMETER For Performance, API, And Regression Testing.

- **User Acceptance Testing (UAT) –**

  Ensures Real-World Usability Before Final Deployment.

Defect Management:

● Bugs Are Classified As Blocker, Major, Minor, And Are Assigned Severity And Priority Levels.

● Developers Fix Defects Based On A Mitigation Table, Followed By Retesting.

Test ID	Test Condition	Expected Output	Actual Output	Remarks
TC_001	User login with valid credentials	Successful login	Successful login	Pass
TC_002	User login with invalid credentials	Error message displayed	Error message displayed	Pass
TC_003	Card scanning using Azure AI	Extracted correct details	Extracted correct details	Pass
TC_004	Contact management – saving a VCard	VCard saved successfully	VCard saved successfully	Pass
TC_005	QR Code & LinkedIn integration	Opens correct LinkedIn profile	Error in redirection	**Fail (Bug reported)**
TC_006	API request to fetch user details	Correct JSON response	Response delay observed	**Fail (Performance Issue)**

❖ TEST RESULTS AND ANALYSIS

❖ Testing Involves Executing Test Cases To Compare Expected And Actual Results. Any Deviations Are Recorded As Defects And Sent For Resolution.

➢ **Testing Process & Bug Resolution**

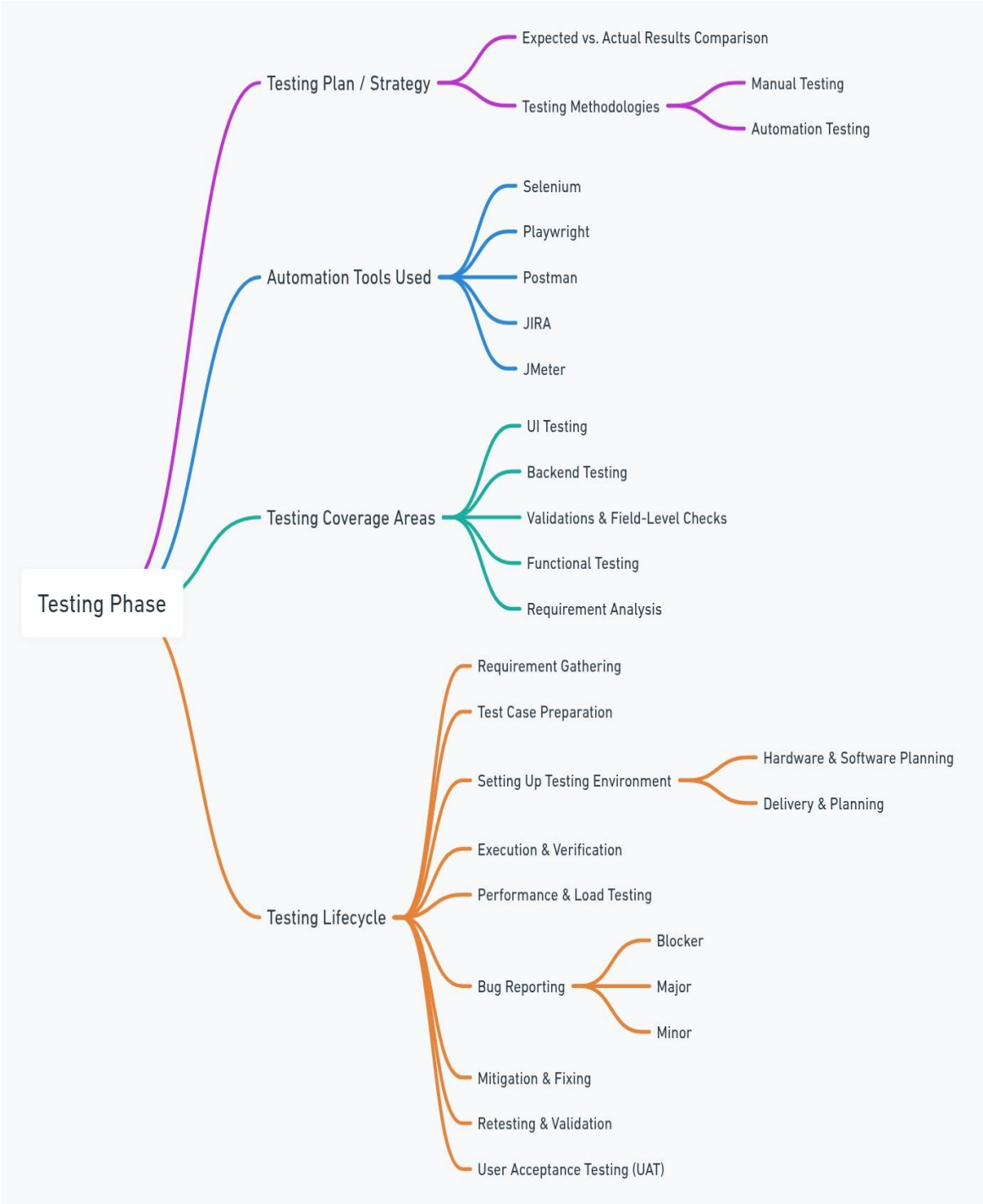
● Test Execution – Run Test Cases And Record Results.

● Bug Reporting – Identify Defects And Assign Severity/Priority.

● Fixing & Retesting – Developers Resolve Issues, Followed By Retesting

● UAT (User Acceptance Testing) – Ensures Final System Approval Before Deployment.

❖ Testing Phase



6 EARNED VALUE AND BURN CHARTS

6.1 PROPOSED ENHANCEMENTS

Future Versions Of The Digital Business Card System For L&T Users May Introduce Several Advanced Features To Enhance Functionality And User Experience.

AI-POWERED CONTACT CATEGORIZATION:

Advanced AI Algorithms Could Categorize Contacts Automatically Based On Their Relationship (E.G., Client, Colleague, Partner) Or Department (E.G., HR, IT, Marketing). This Would Enable Users To Quickly Sort And Access Contacts Based On Context, Improving Organizational Efficiency.

ENHANCED NETWORKING ANALYTICS:

The System Could Include Analytics Tools That Offer Insights Into Networking Opportunities. Features Could Include Identifying Key Contacts For Collaboration, Tracking Interaction Frequencies, And Recommending New Connections Within The Organization Based On Shared Projects Or Interests.

INTEGRATION WITH INTERNAL COMMUNICATION TOOLS:

Seamless Integration With L&T's Internal Communication Platforms, Such As Microsoft Teams Or Share point, Would Allow Users To Easily Share Business Cards Within Chats, Emails, And Calendar Events. This Integration Would Streamline Communication And Collaboration Efforts By Providing Instant Access To Contact Details Within The Tools Employees Already Use.

These Proposed Enhancements Aim To Provide Smarter, More Intuitive Contact Management, Ultimately Fostering Stronger Networking And Collaboration Within L&T, And Improving Productivity Across The Organization.

CONCLUSION

The Digital VCard System, developed as part of my internship, is an innovative web and mobile application designed to digitally share professional contact details without the need for a physical business card. Built using .NET, Azure AI, JavaScript, HTML, CSS, FIGMA, APIs, and SQL, the system follows the MVC architecture and the Waterfall software development lifecycle. The project includes key features such as secure authentication via employee PS number, automatic data retrieval from the company database, Azure AI powered card scanning, contact management, LinkedIn integration, QR code generation, and direct call, email, and messaging options. Each module was carefully designed, tested, and integrated to ensure smooth functionality, efficiency, and user-friendliness.

7 SCREEN SHOTS OF PROJECT

Screen Shots before Login

♣ LOGIN :

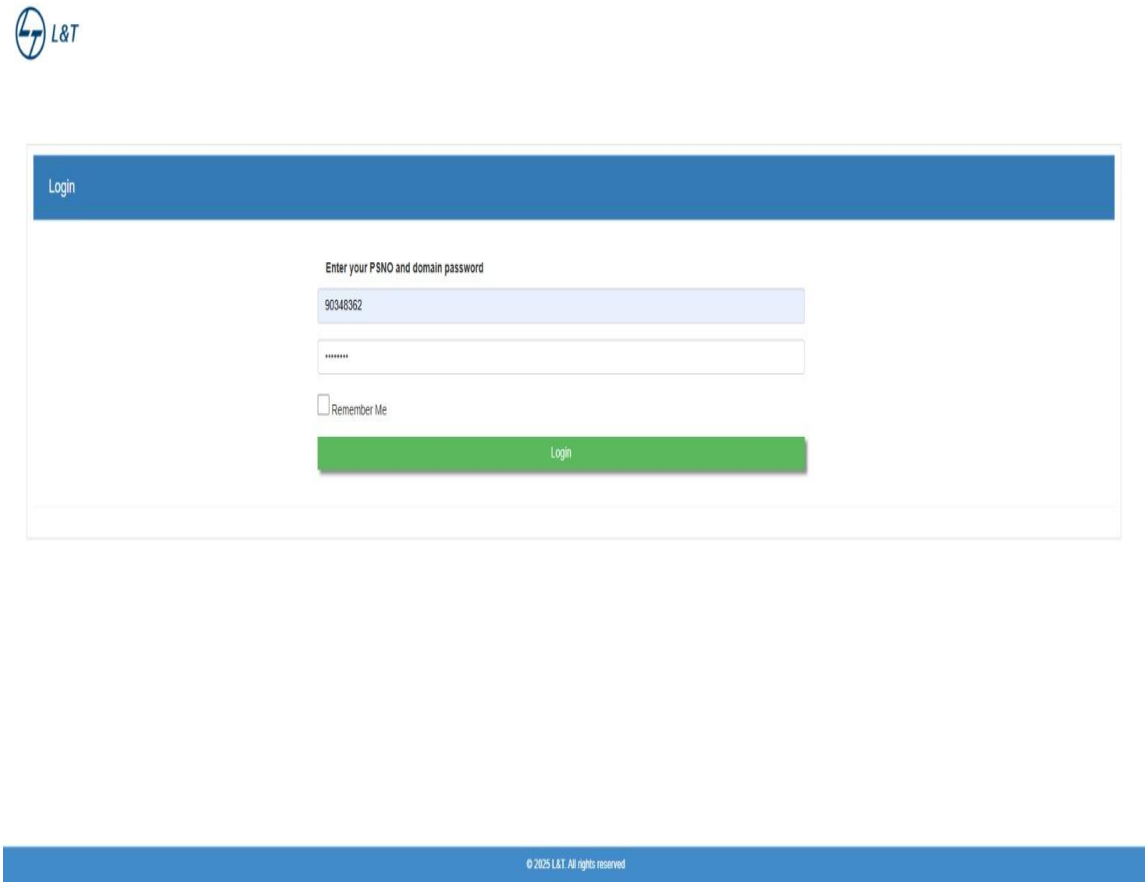


FIGURE 1 : USER AUTHENTICATION

Login :

- L&T id
- Password
- Create account

PROFILE PAGE :

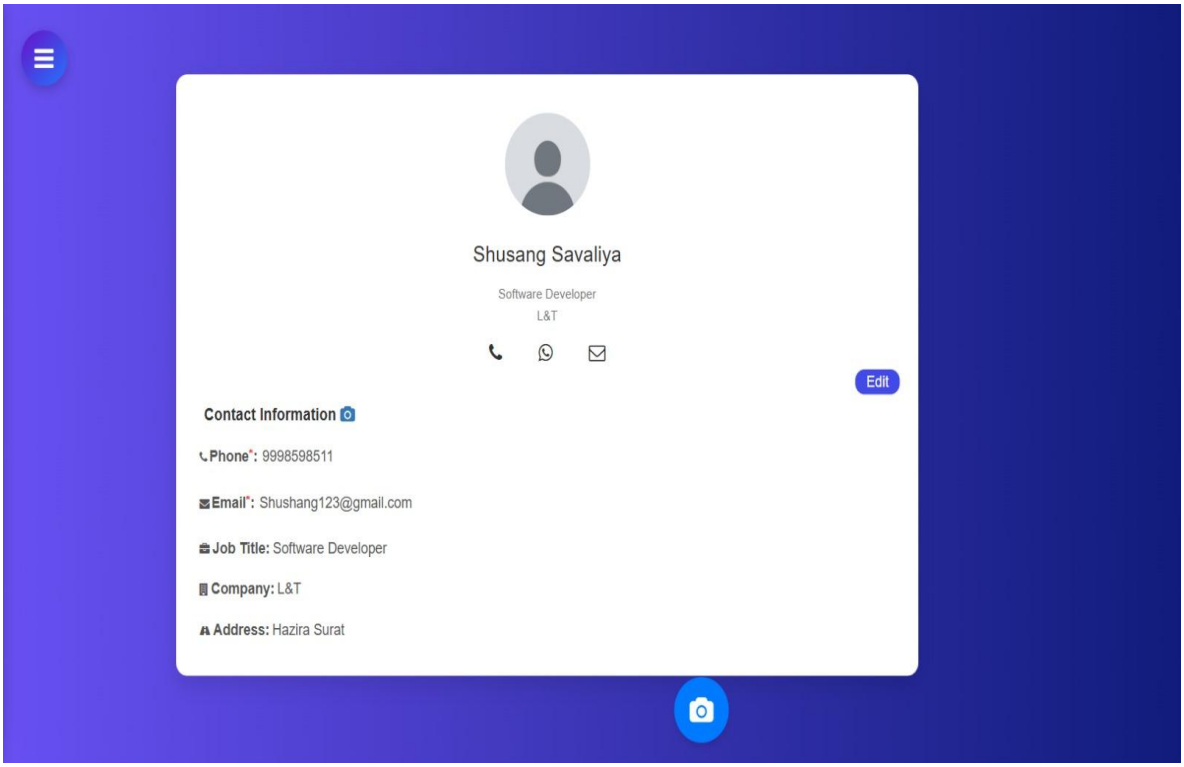


FIGURE 2 : PROFILE

Profile Page :

- It will be open profile page
- Home
- Menu
- Email
- company
- Mobile no
- Address

♣ EDIT PROFILE:

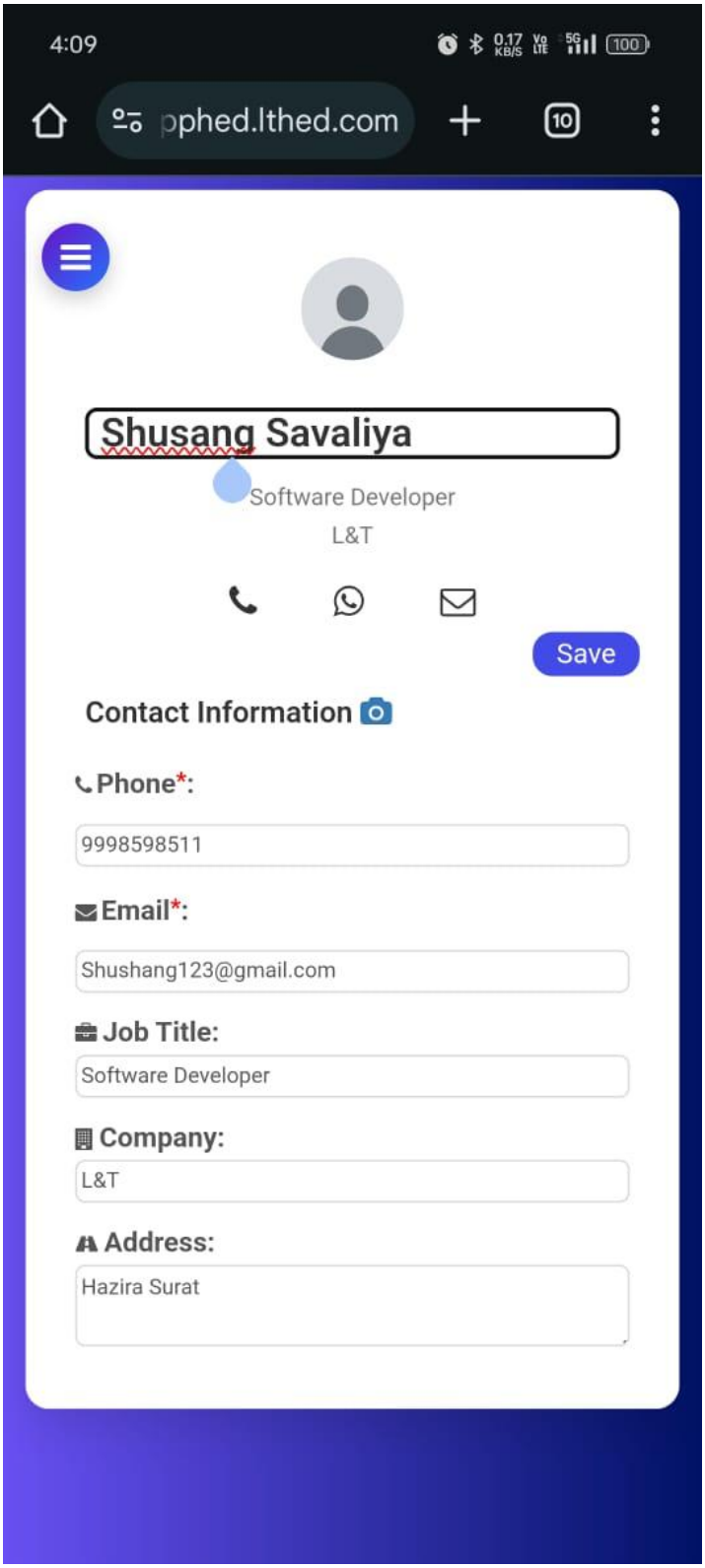


FIGURE 3 : EDIT PROFILE

- **QR CODE(URL) :**



FIGURE 5 :QR CODE(URL)

♣ **QR CODE(card) :**

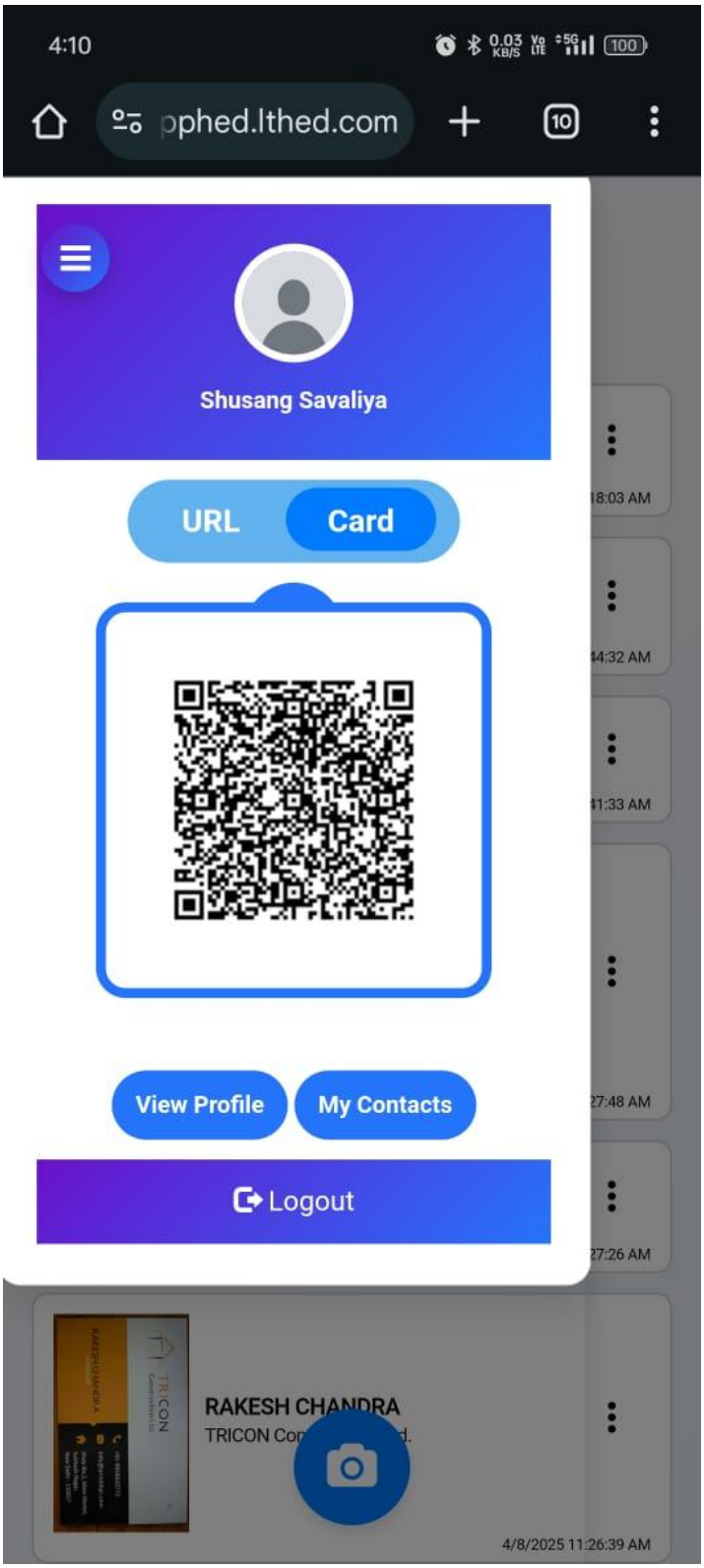


FIGURE 5 :QR CODE(CART)

♣ CAMERA MODULE

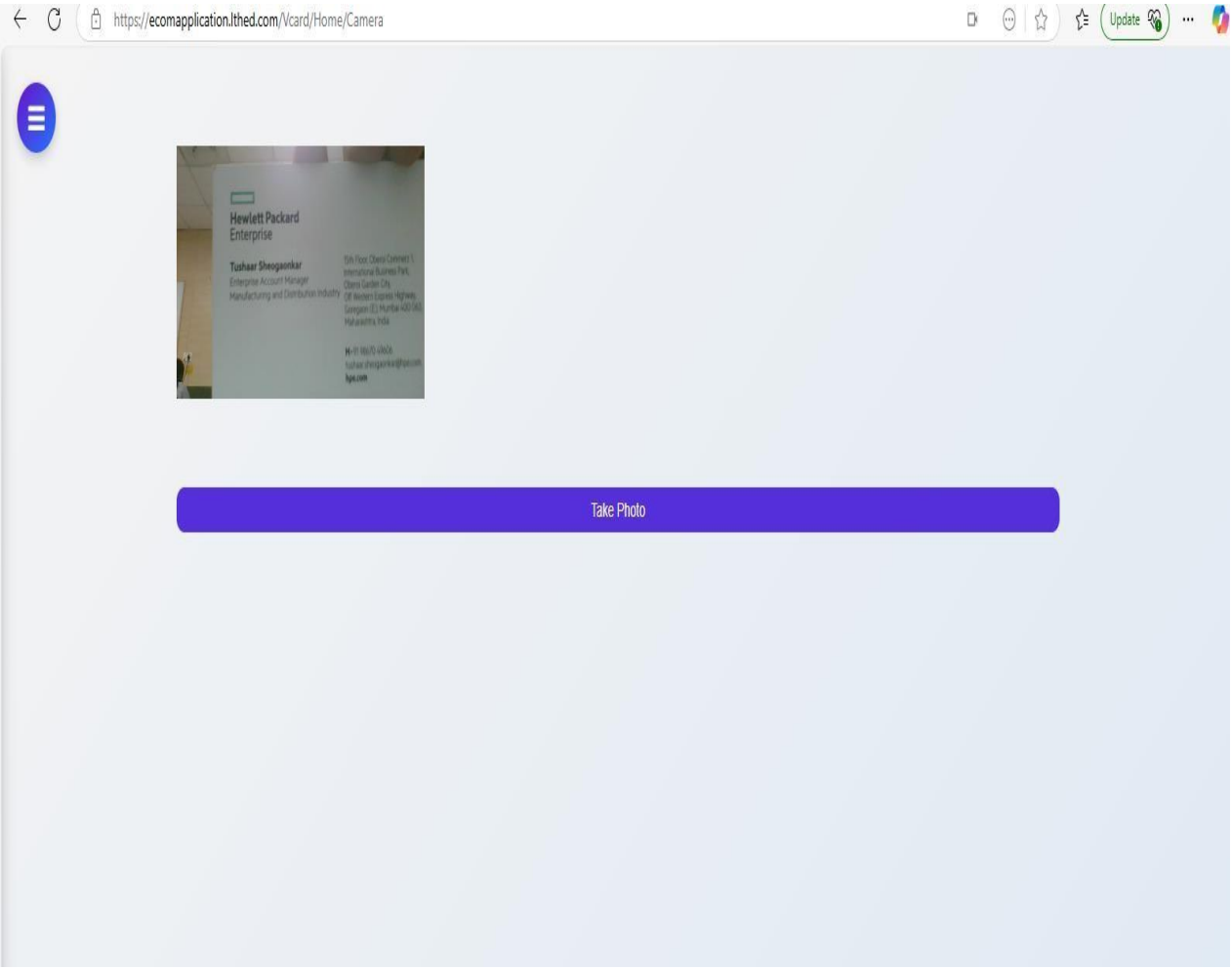


FIGURE 6 : CAMERA MODULE

♣ PROFILE THROUGH AZURE

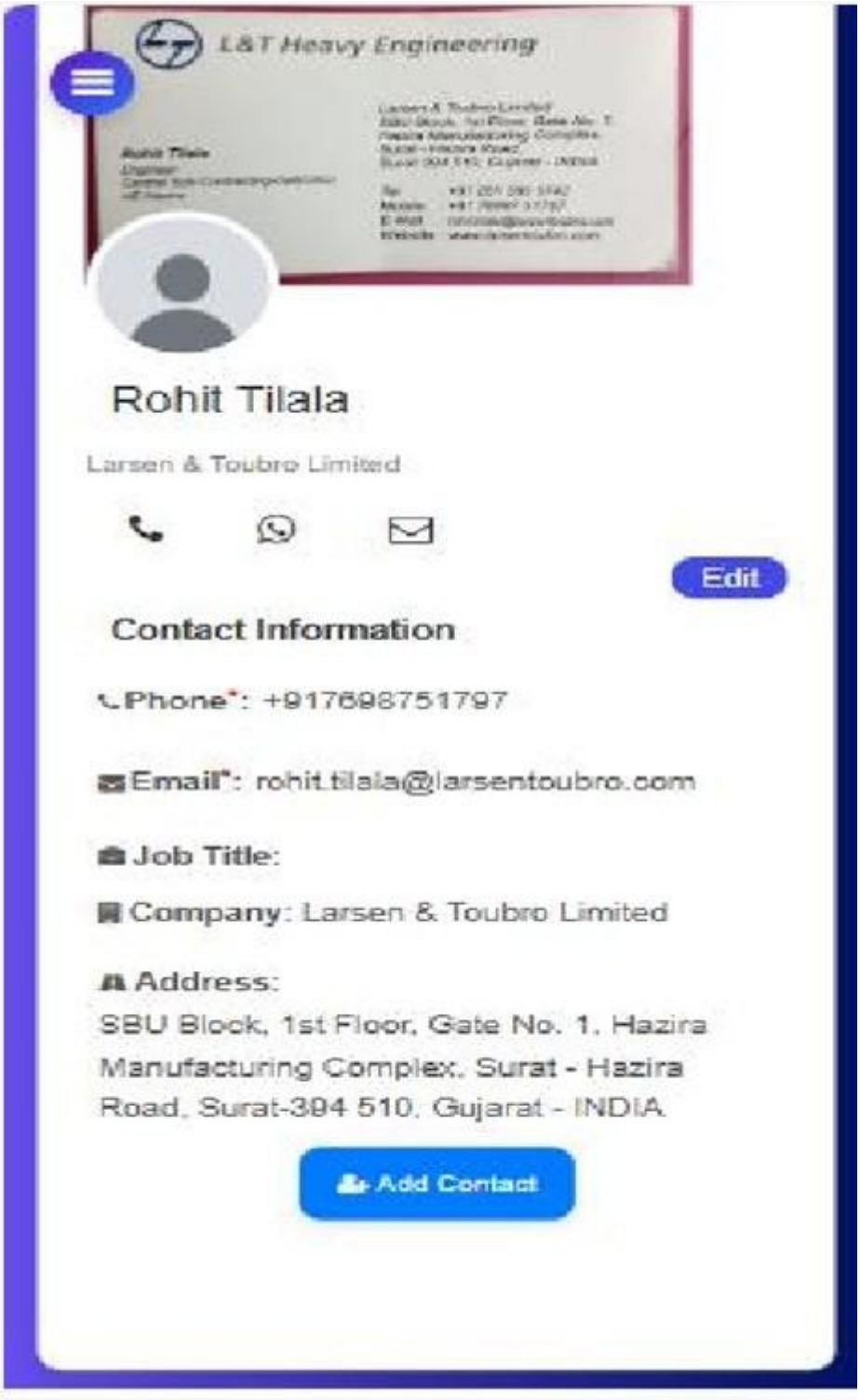


FIGURE 7:PROFILE THROUGH AZURE

♣ CONTACT MANAGEMENTAI SERVICES

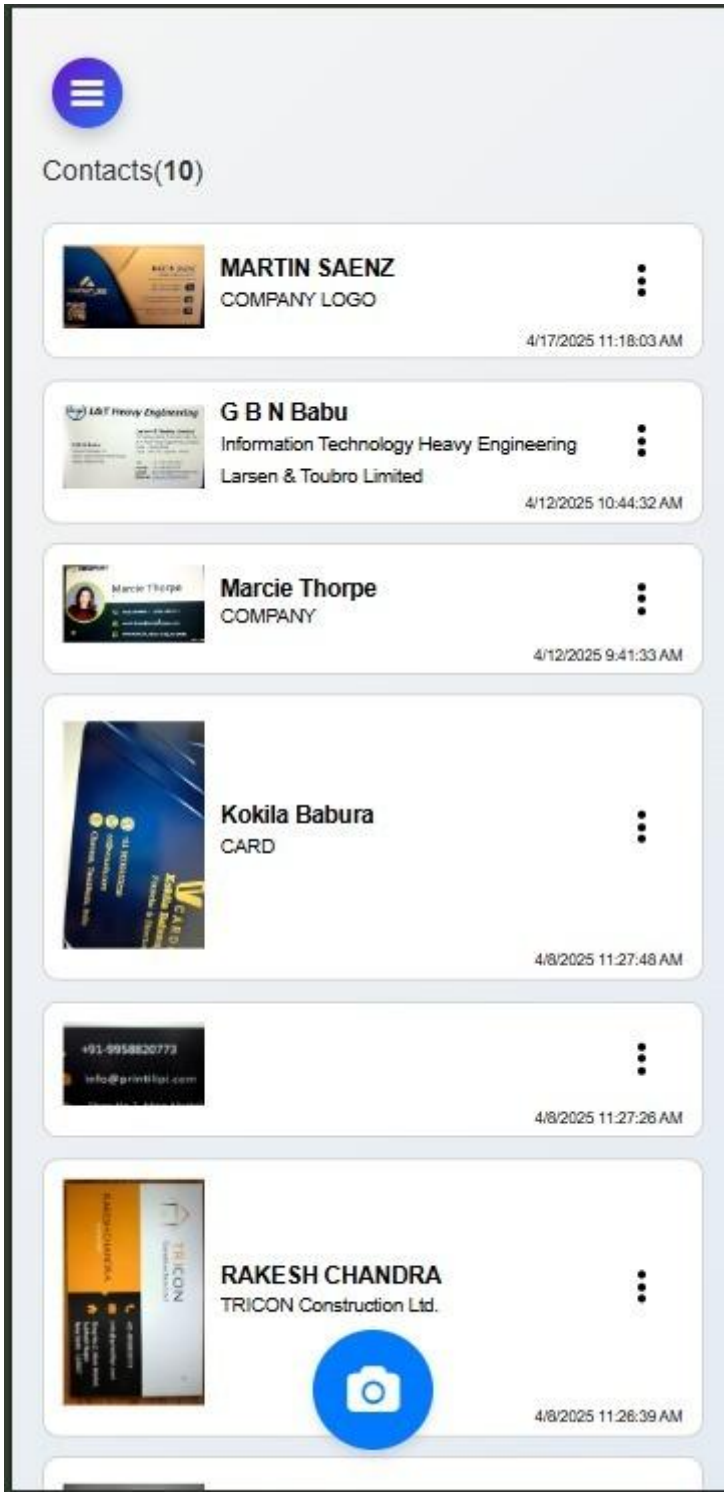


FIGURE 8 : CONTACT MANAGEMENT AI SERVICES

♣ MANAGE CARD

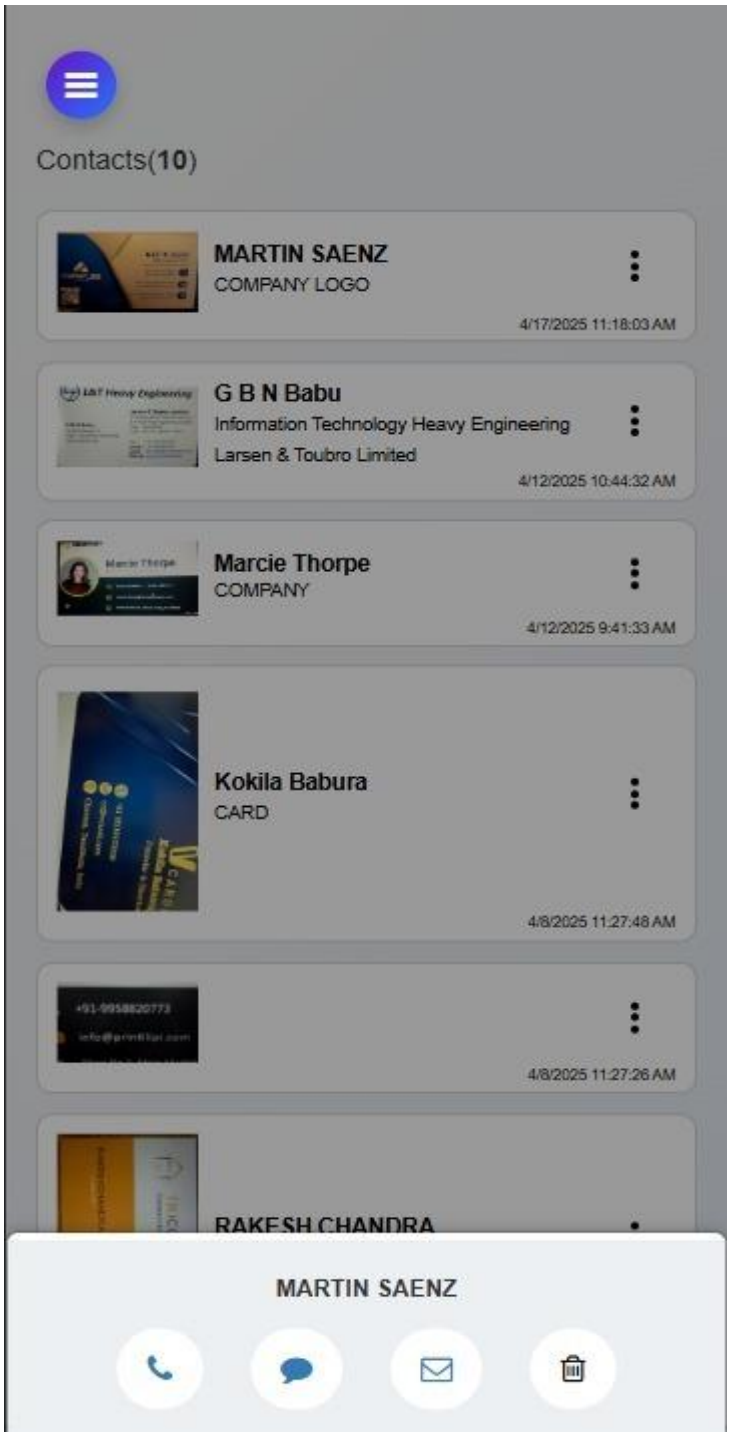


FIGURE 9 : MANAGE CARD

♣ DELETE CARD

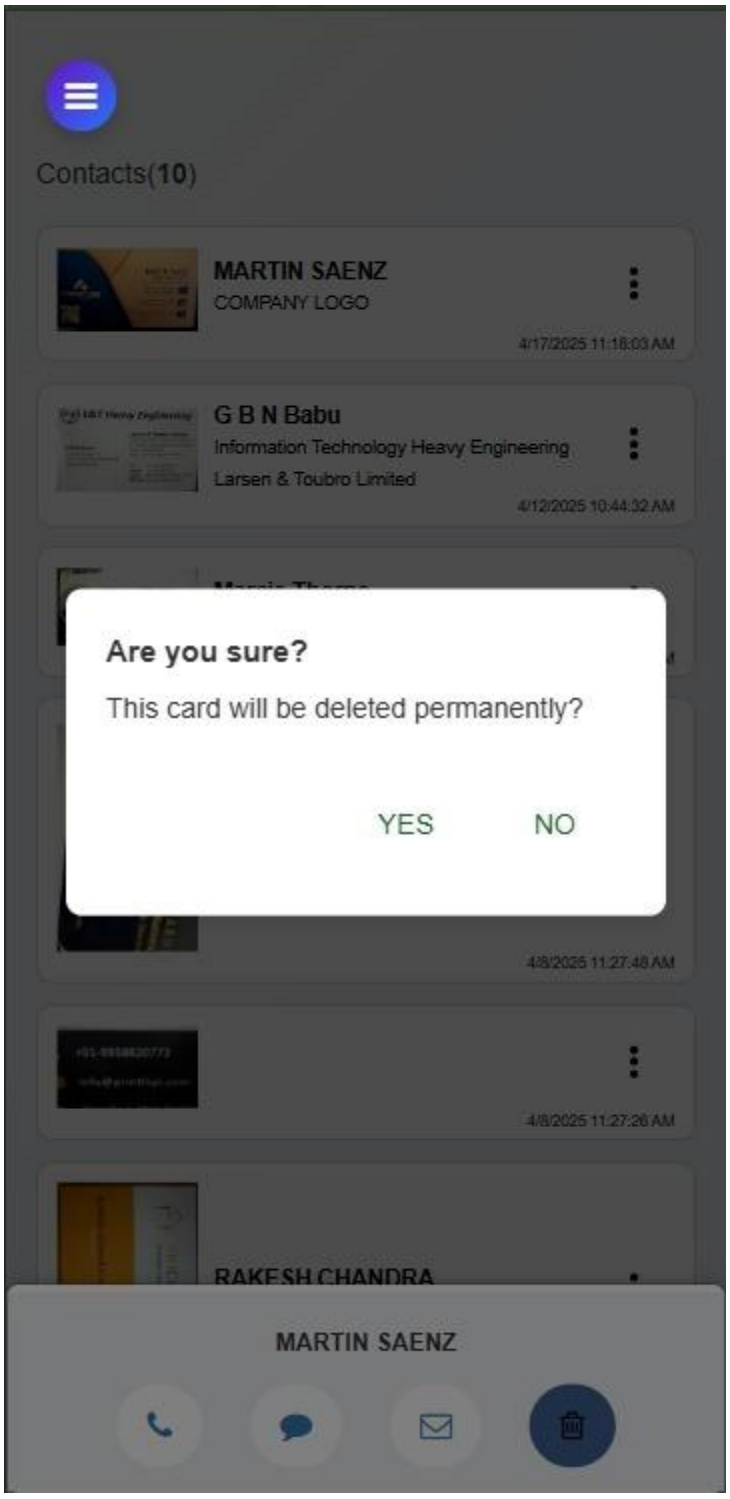


FIGURE 8 : DELETE CARD

❖ REFERENCE

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• Azure DEVOPS Documentation: https://learn.microsoft.com/en-us/azure/devops/